
ABIMS: a New Tool to Assess Measurement Inconsistencies in Bilateral International Migrants Stocks Databases

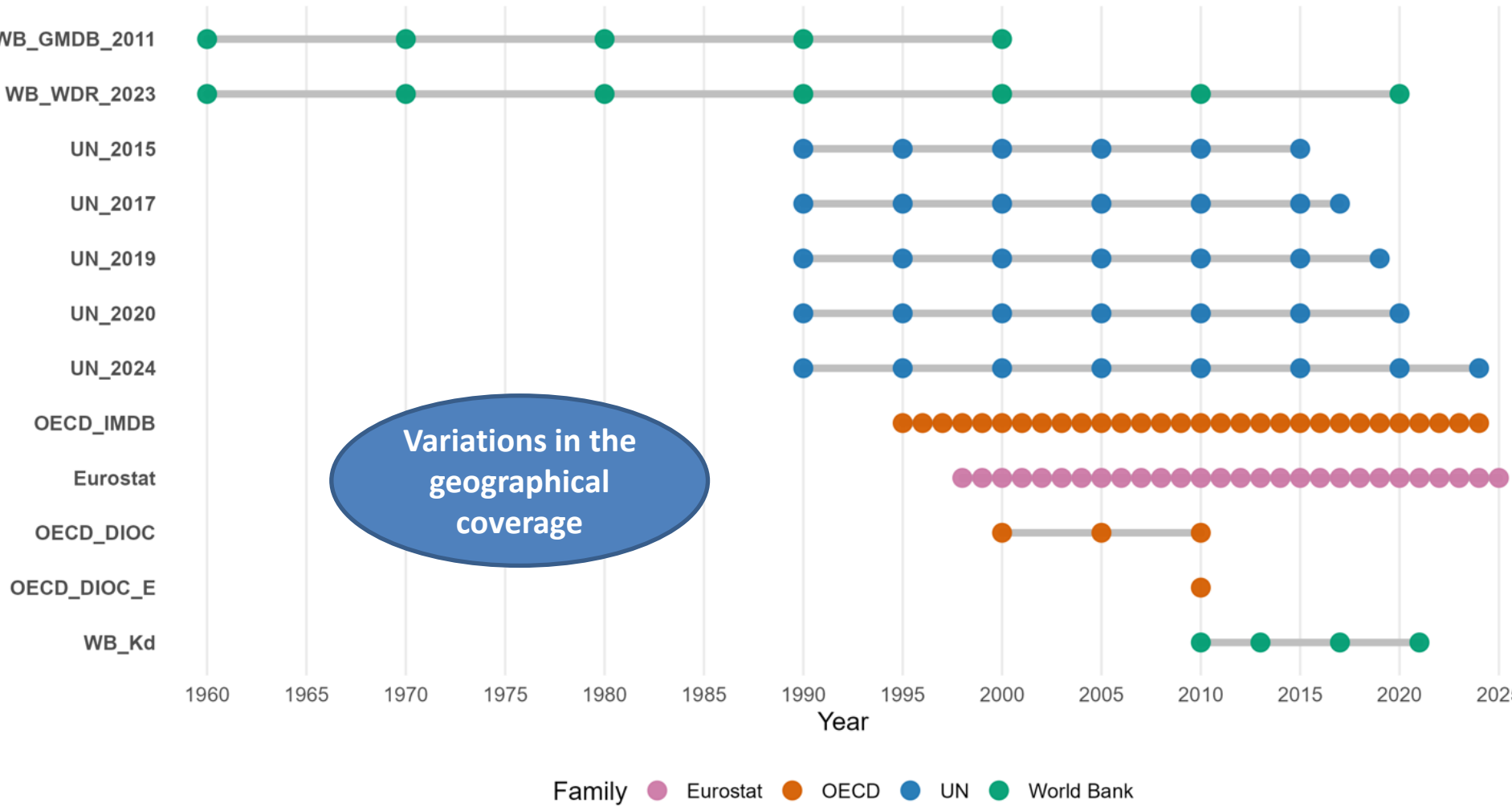
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Bruno Schoumaker (UCL)
Patrick N'twali (Ined, UCL)

Introduction

- **BIMS-DB: *Bilateral International Migrants Stocks Databases***
 - Produced by international organizations: WB, UN, OECD
 - Origin-destination matrices (country pairs)
 - Aggregation of national sources
- **A revolution in migration measurement in the 2010s**
>>> major sources for...
 - Policy reports and migration research
 - Analyses of trends: immigrant and emigrant stocks
 - Estimations of flows at the global level
 - Studies on the causes and consequences of migration

A stunning profusion of BIMS-DB...

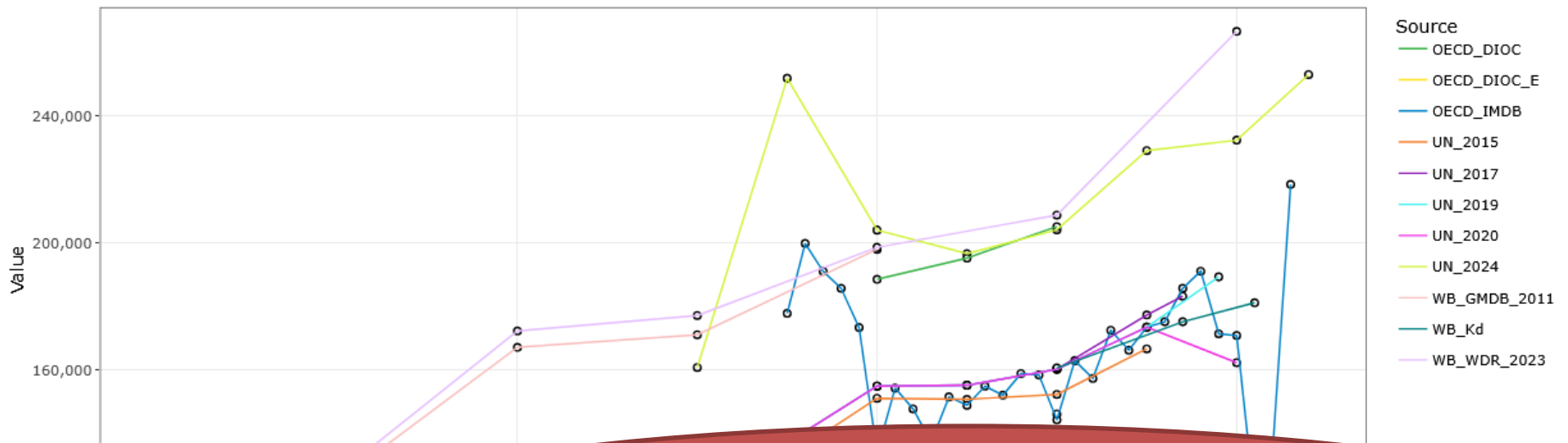
BIMS-DB - Time coverage



... with serious inconsistencies

French migrants in the USA

Outlier Detection: FRA → USA



- What is the prevalence of these inconsistencies?
 - What are the reasons?
 - What are the impacts?



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In search of a systematic detection of outliers

Methods

1. Assemble all BIMS-DB over 1960-2024

>>> Aggregated BIMS-DB (ABIMS-DB)

- A huge work to homogeneize the BIMS-DB
 - files retrieval (disappearance, revisions)
 - files formats and contents, country codes
- More than 31,000 country pairs, 770 000 observations

2. For the migrant stock of each country-pair, model a trend (1960-2024) using BIMS-DB observations

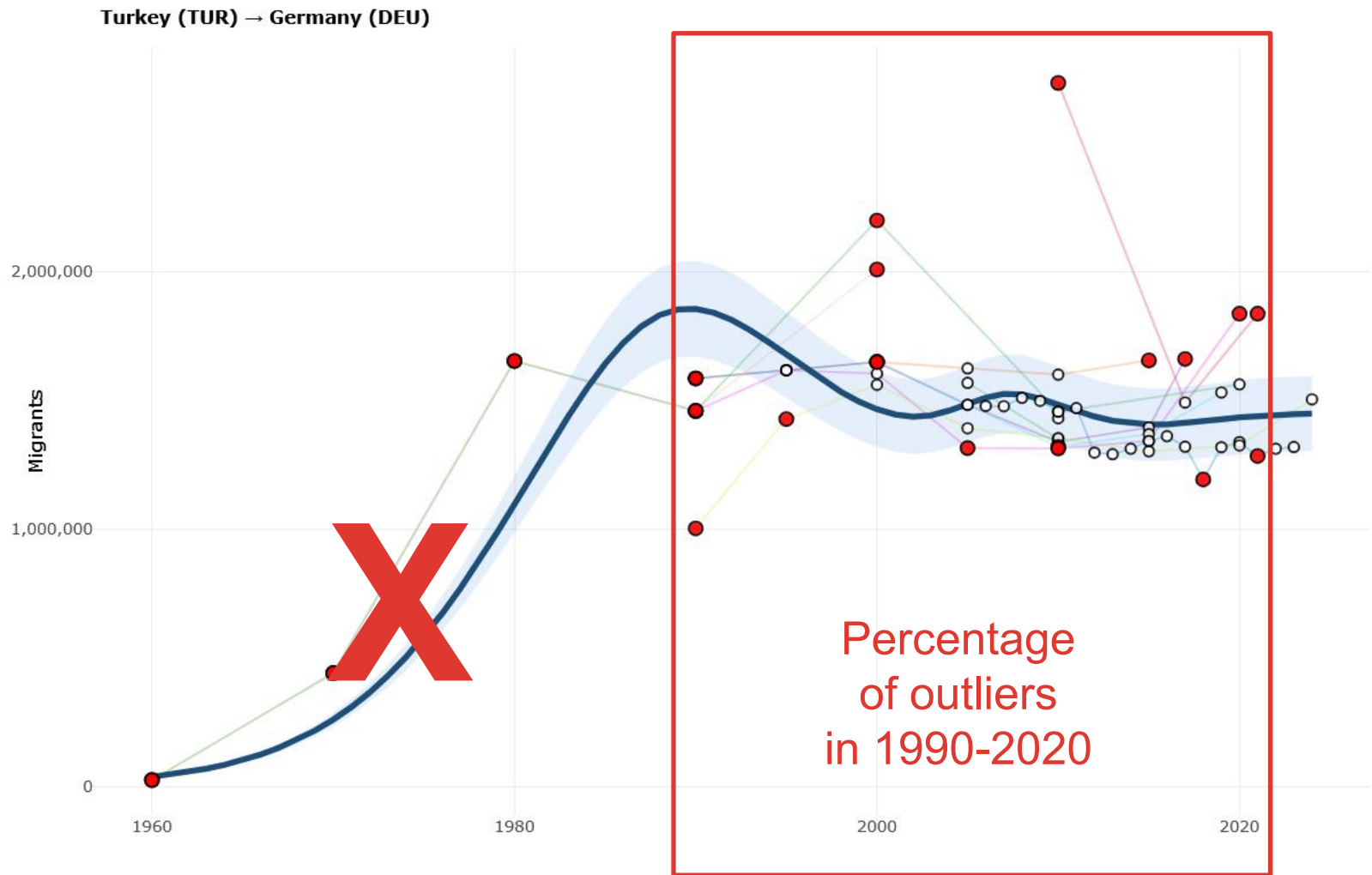
- Log-gam Model
- Automatic smoothing of the spline
- A minimum of 8 observations, at least 2 data sources >>> 12,261 pairs

3. Measure the gap between observed and predicted values

- *Ad hoc* tolerance band: relative variation of +/- 10% from the true value
 - 10% threshold necessarily arbitrary
- Indicator: % of outliers in each country pair in 1990-2024

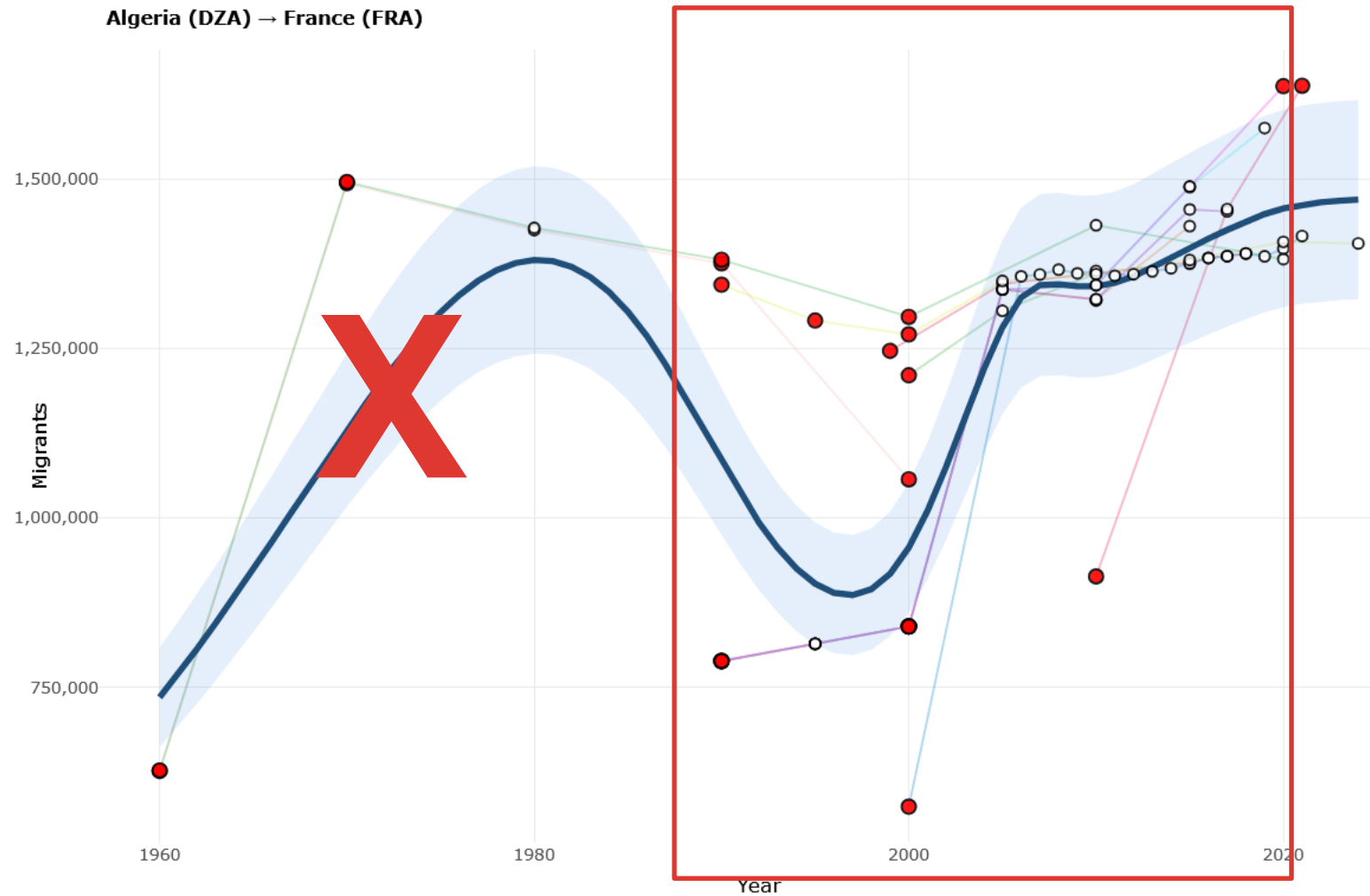
A work in
progress!

Turkish migrants in Germany



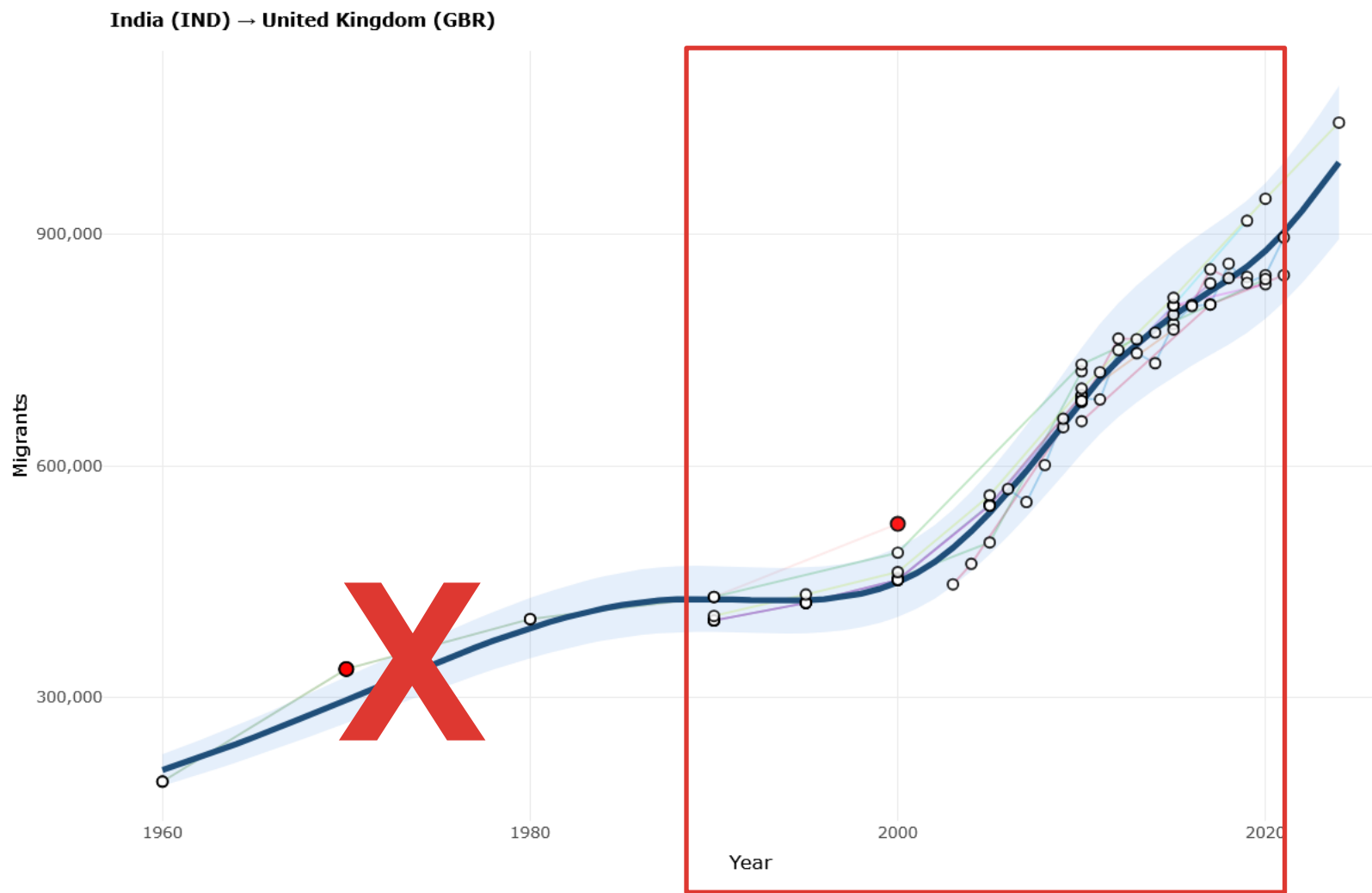
Blue area (tolerance band): +/- 10% around the modelled spline
Red dots: outliers

Algerian migrants in France



Blue area (tolerance band): +/- 10% around the modelled spline
Red dots: outliers

Indian migrants in the UK

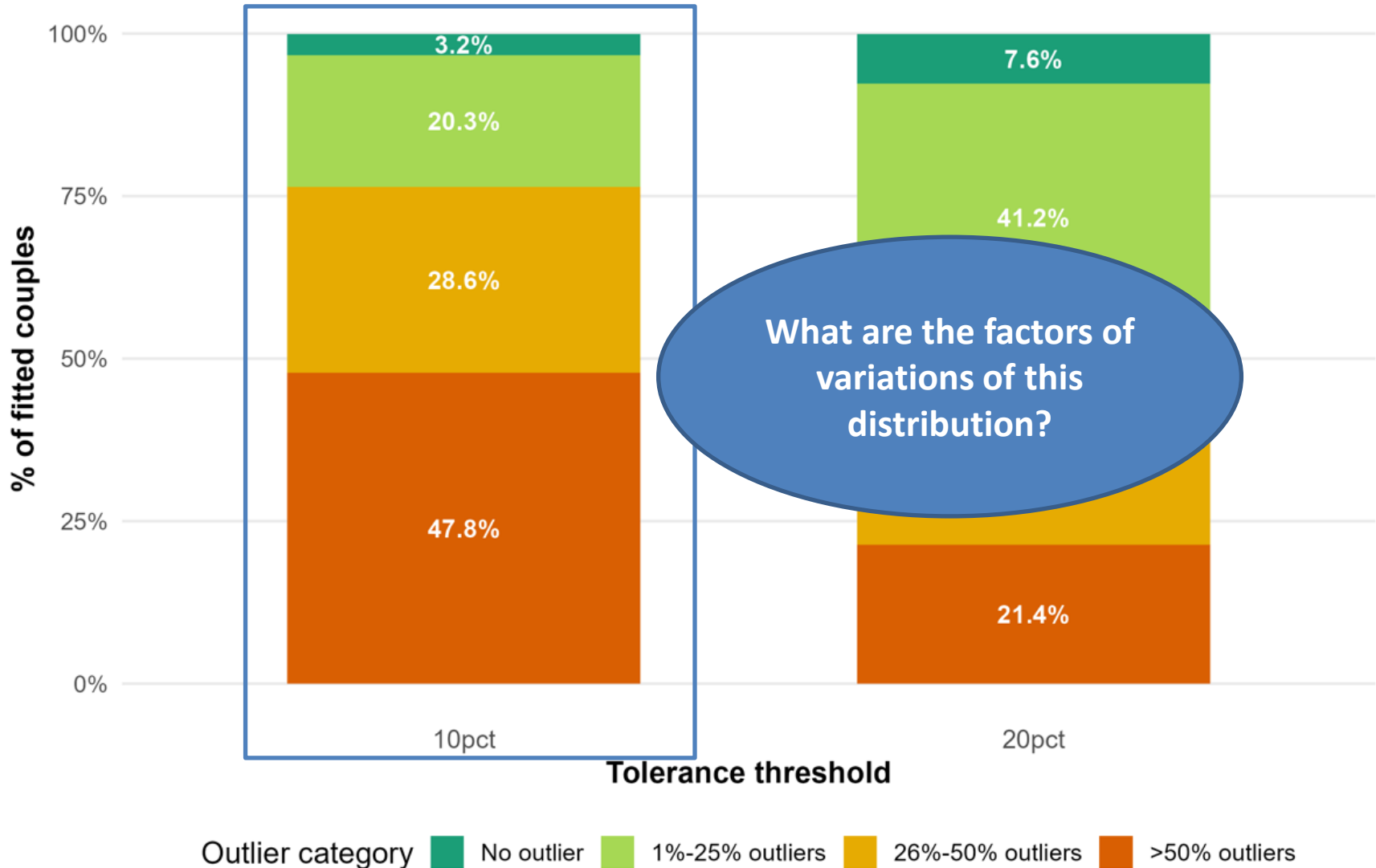


Blue area (tolerance band): +/- 10% around the modelled spline
Red dots: outliers

Percentage of country pairs with outliers

Log-GAM ad hoc tolerance band: comparison of +/-10% and +/-20%

N = 12,261 couples with fitted model





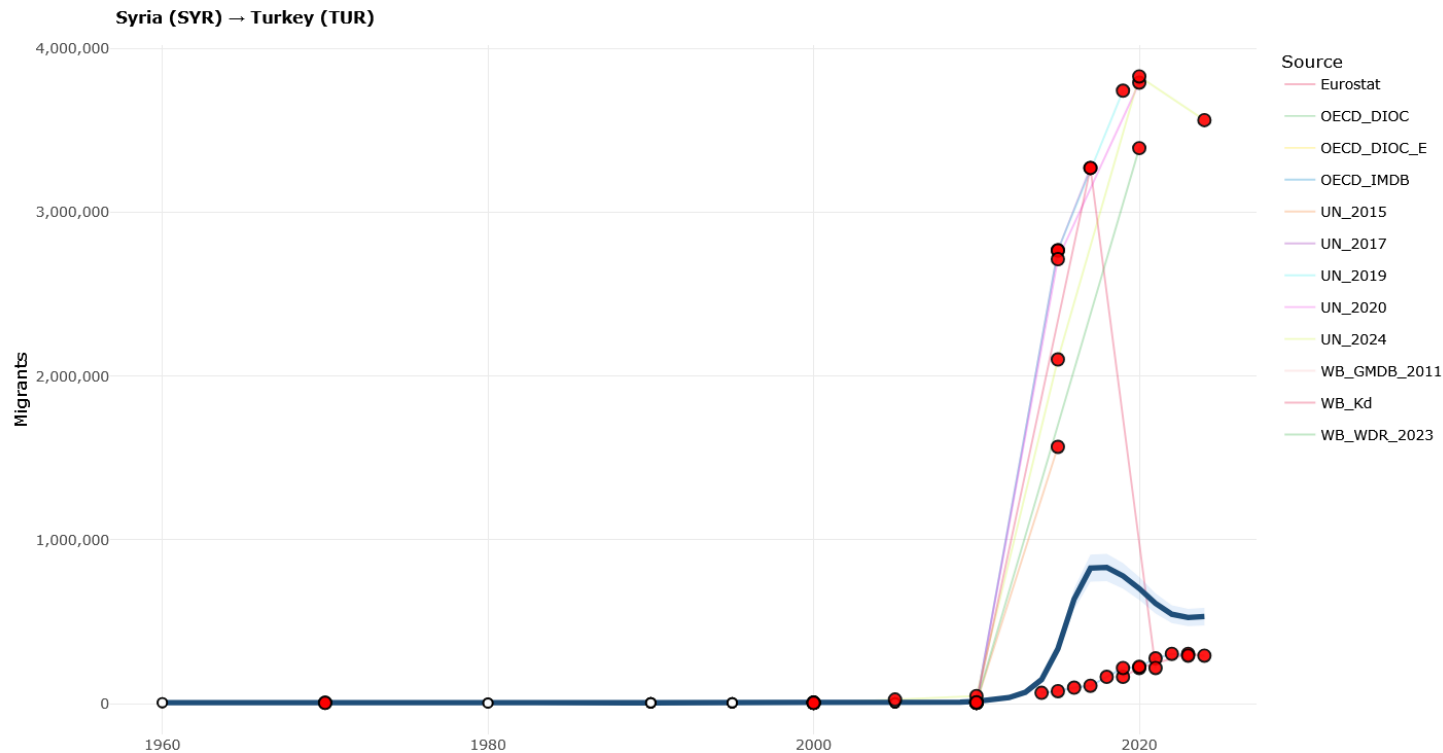
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An exploration of the factors associated to discrepancies in migrant stocks estimates (1990-2020)

Many possible reasons for differences in BIMS-DB estimates

- **Different migrants definition**
 - **Foreign born** vs. **foreigner** >>> differences in data sources; adjustments *or not*
 - Adding **refugees** (from UNHCR) to national stocks *or not*



Many possible reasons for differences in BIMS-DB estimates

- **Different primary data sources**
 - Censuses *or* other sources
 - Obtained through National statistical offices (NSOs) or adapted from the UN

- **Different methods in case of missing information from NSOs**
 - >>> imputations *or not*
 - >>> Hypotheses: more discrepancies in estimates when...
 1. census data is lacking

SURPRISE! The quantity of outliers is poorly associated with the wealth of national data sources

Outlier Share by Census Group — Destination

census count classes, 1970–2020



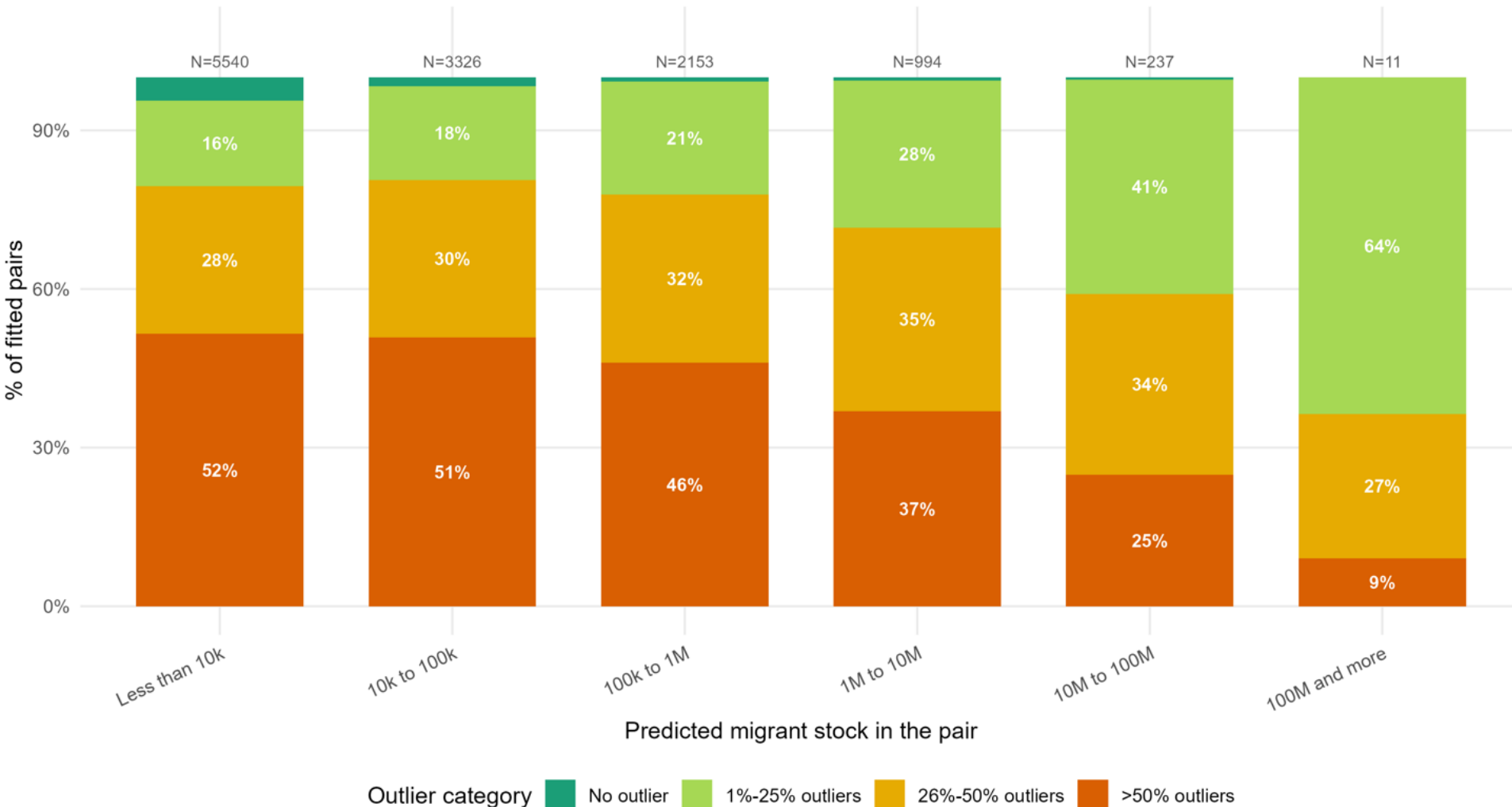
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 - >>> imputations *or not*
 - >>> Hypotheses: more discrepancies in estimates when...
 1. census data is lacking [census availability]
 2. the migrant stock of a country-pair is small, because NSOs are reluctant to deliver the data [census accessibility]

INDEED! Larger country-pairs are much less subject to discrepancies

Outliers by Predicted Migrant Stock

Log-GAM +/-10%; stock = sum of trend_pred within pair



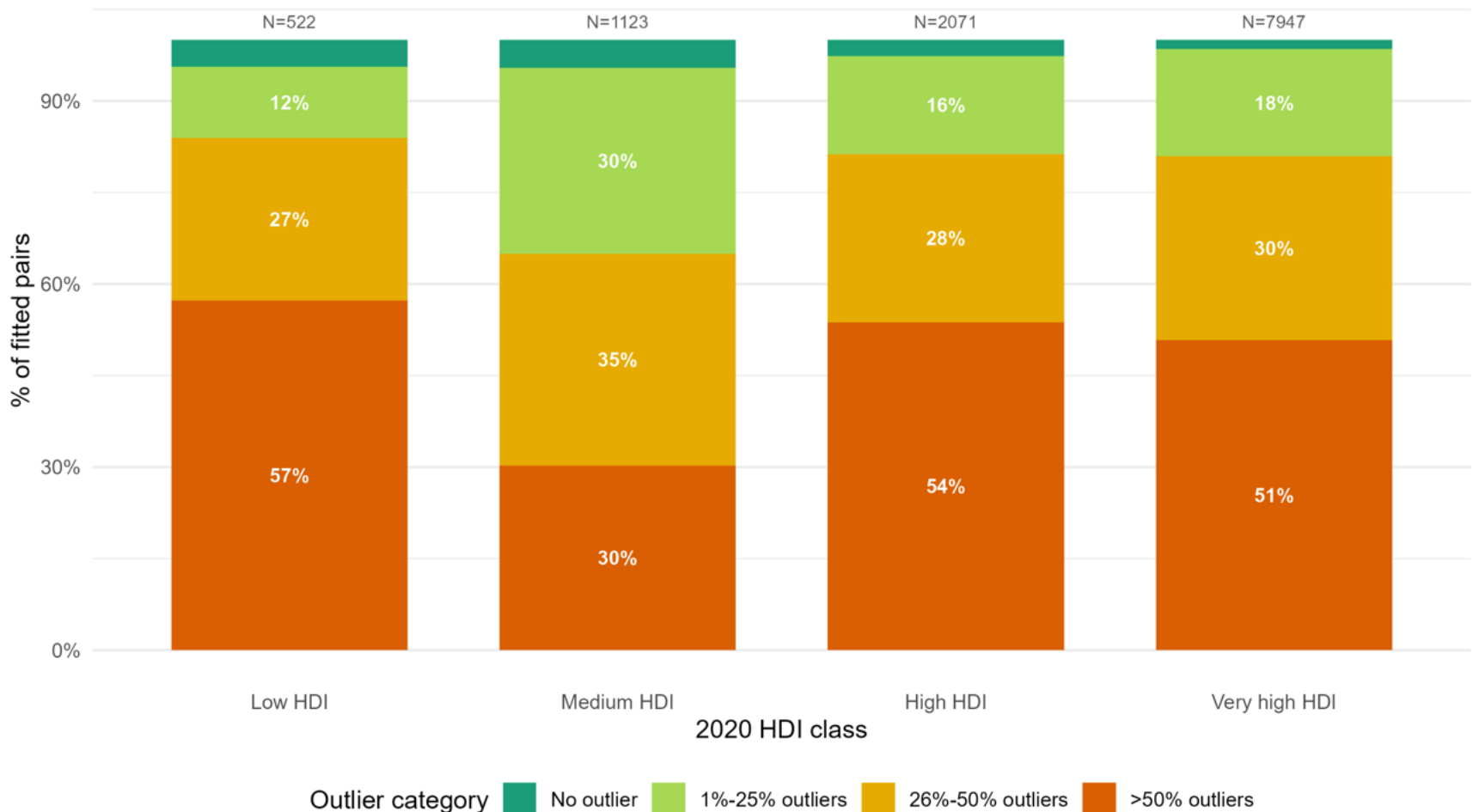
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 1. census data is lacking [census existence]
 2. the migrant stock of a country-pair is small, because NSOs are reluctant to deliver the data [census accessibility]
- **Human mistakes in the data collection system**, from NSOs to international organisations
 - >>> Hypothesis: NSOs in less developed countries have less human resources to fill in international organisations requests

NOT REALLY! The quantity of outliers is not clearly associated to the developement level of the destination country

Outliers by 2020 HDI - Destination

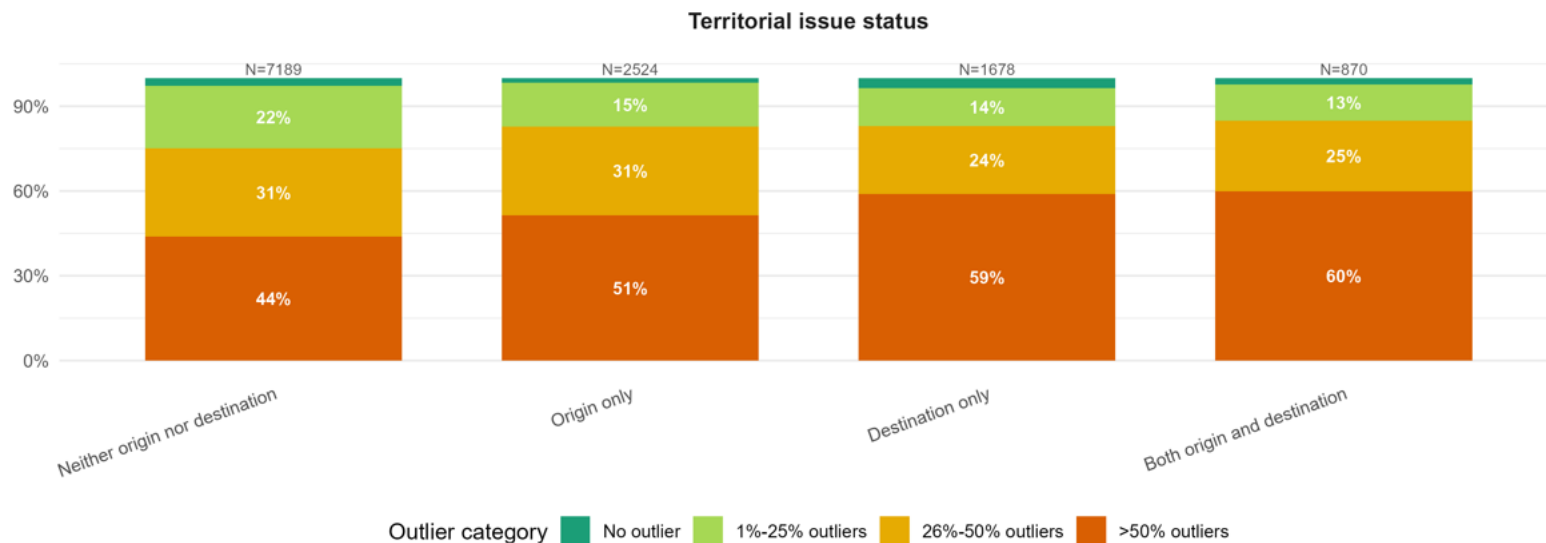
HDI classes; N = number of fitted pairs



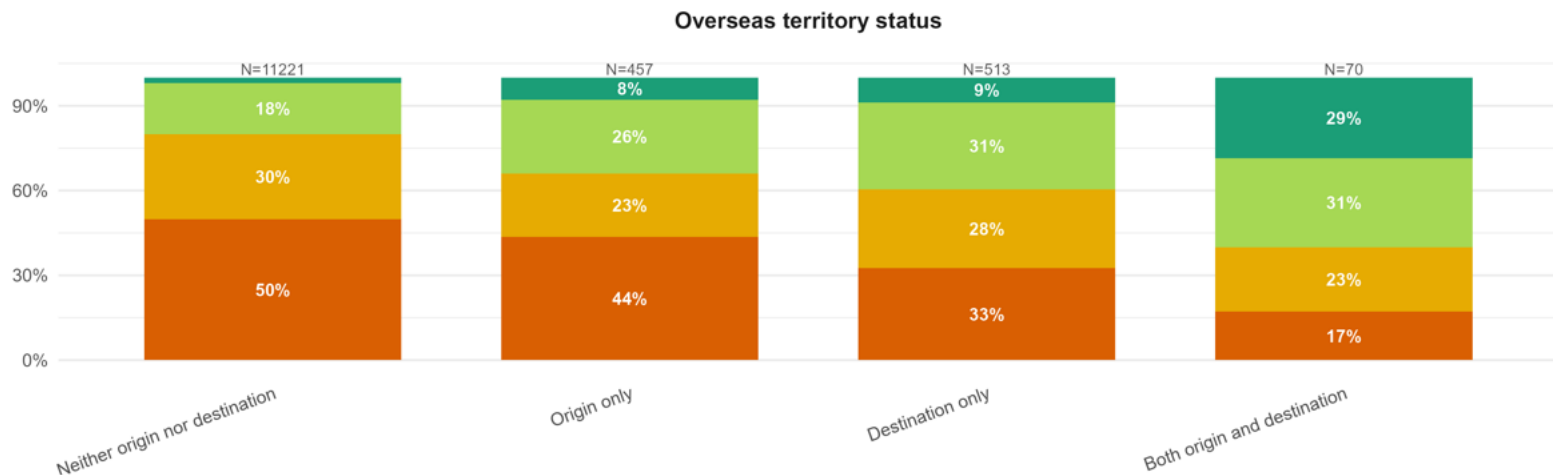
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- **Geographical coverage**
 1. Territorial recompositions over time >>> imputations *or not*
 - E.g. USSR or Yugoslavia dissolution, decolonization...
 2. Overseas territories are counted with their metropolis *or not* ; variations over time

INDEED! Border issues create discrepancies in estimates



BIG SURPRISE! Overseas territories are less subject to discrepancies





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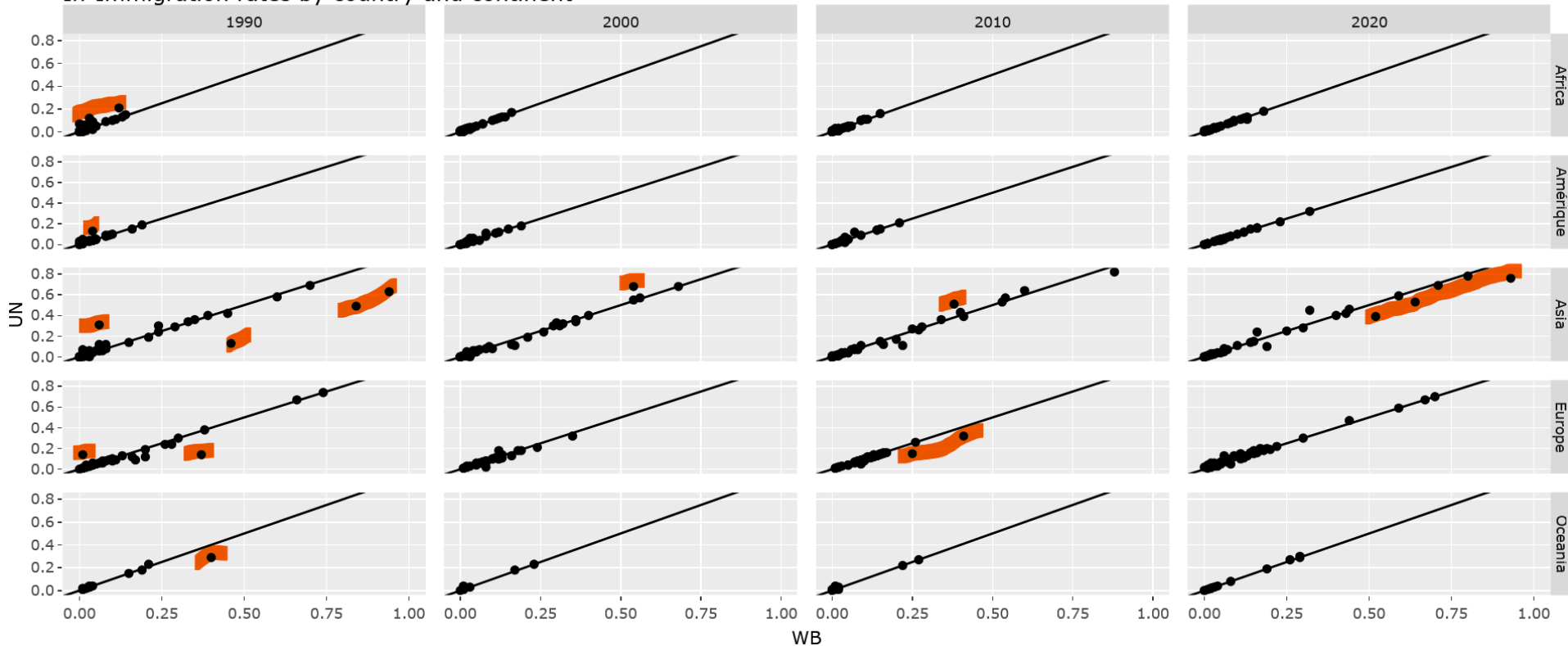
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Impacts on the measurement of migration trends World Bank 2023 vs. United Nations 2024

Country level, by continent and date

Percentage of **immigrants** in destination countries (1990-2020)

In-Immigration rates by country and continent

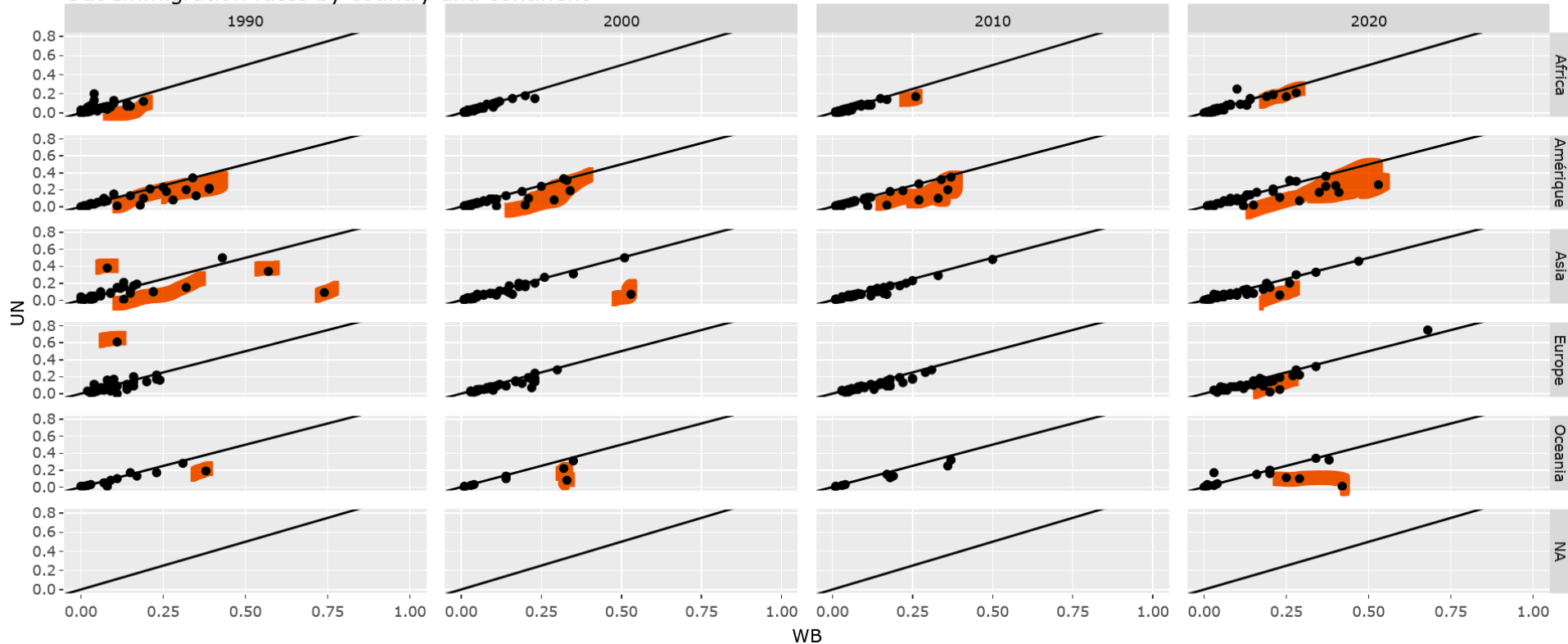


- Gaps between UN and WB diminish over time
 - Discrepancies in small territories
 - Especially in the Middle East

Country level, by continent and date

Percentage of **out-migrants**
compared to the population born in the origin country (1990-2020)

Out-Immigration rates by country and continent



- Gaps persist over time
- Also concentrated in small territories
 - But scattered in all continents

Conclusion



Conclusion

- Summary results
 - A significant prevalence of inconsistencies
 - Expected reasons for discrepancies are not fully confirmed
 - Overseas are less problematic than other territories (?)
 - No relationship with data wealth (number of censuses, HDI)
 - Impact:
 - Discrepancies affect marginally national migration indicators
 - Concentration of issues in small countries
 - >>> Discrepancies have mainly an impact at the country-pair level
- Further research needed
 - To better identify outliers
 - other sorts of models and tolerance bands
 - Pairs with only one source
 - To deepen the understanding of the causes of discrepancies
 - Other factors to be explored: refugees...
 - ABIMS-DB in open access, including with new « clean » estimates
 - To assess impacts (flows estimates, determinants...)
 - To work with institutions to improve the BIMS-DB production process



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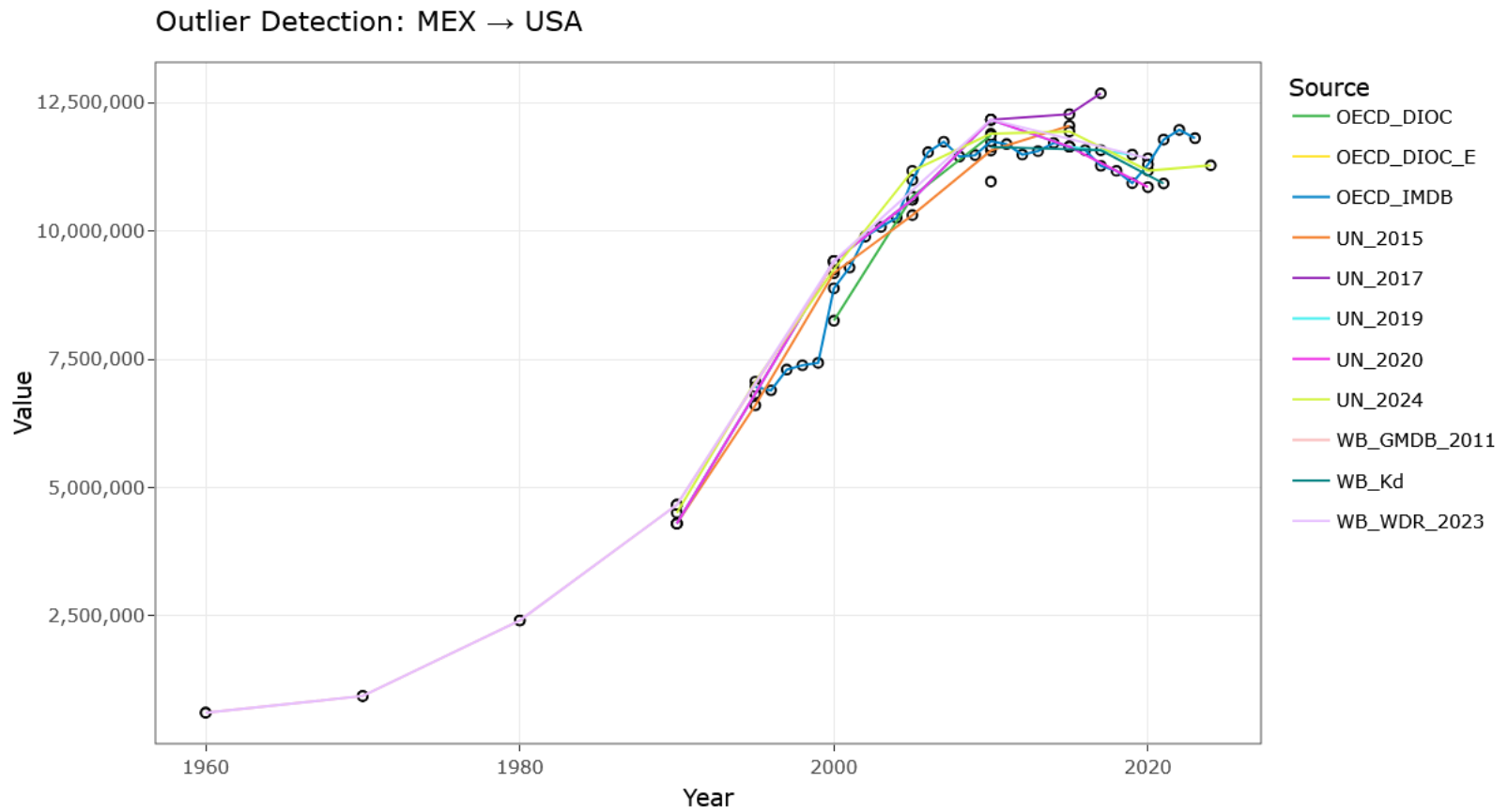
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Additional material

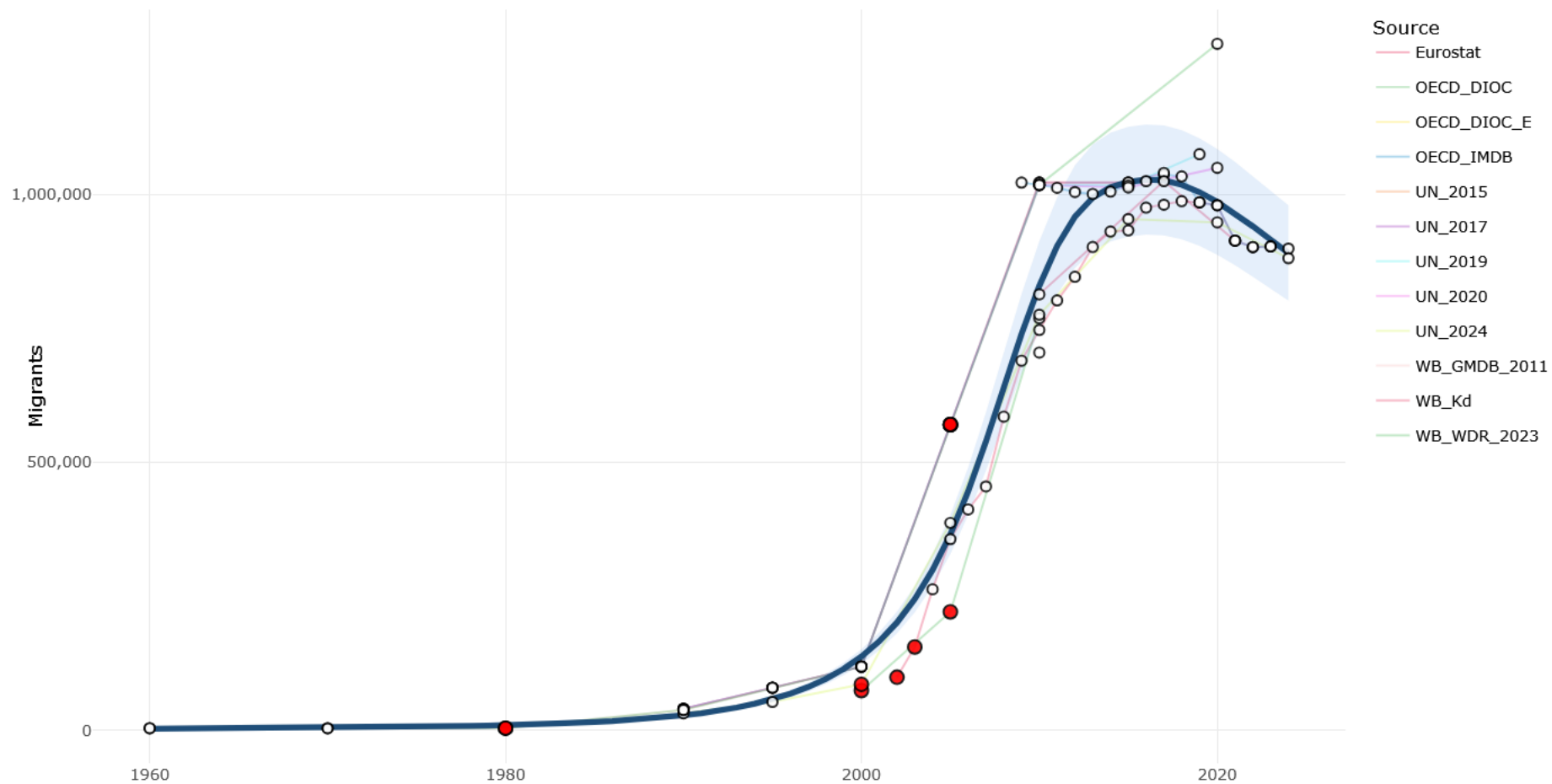
Des cas de bonne convergence des BIMS-DB



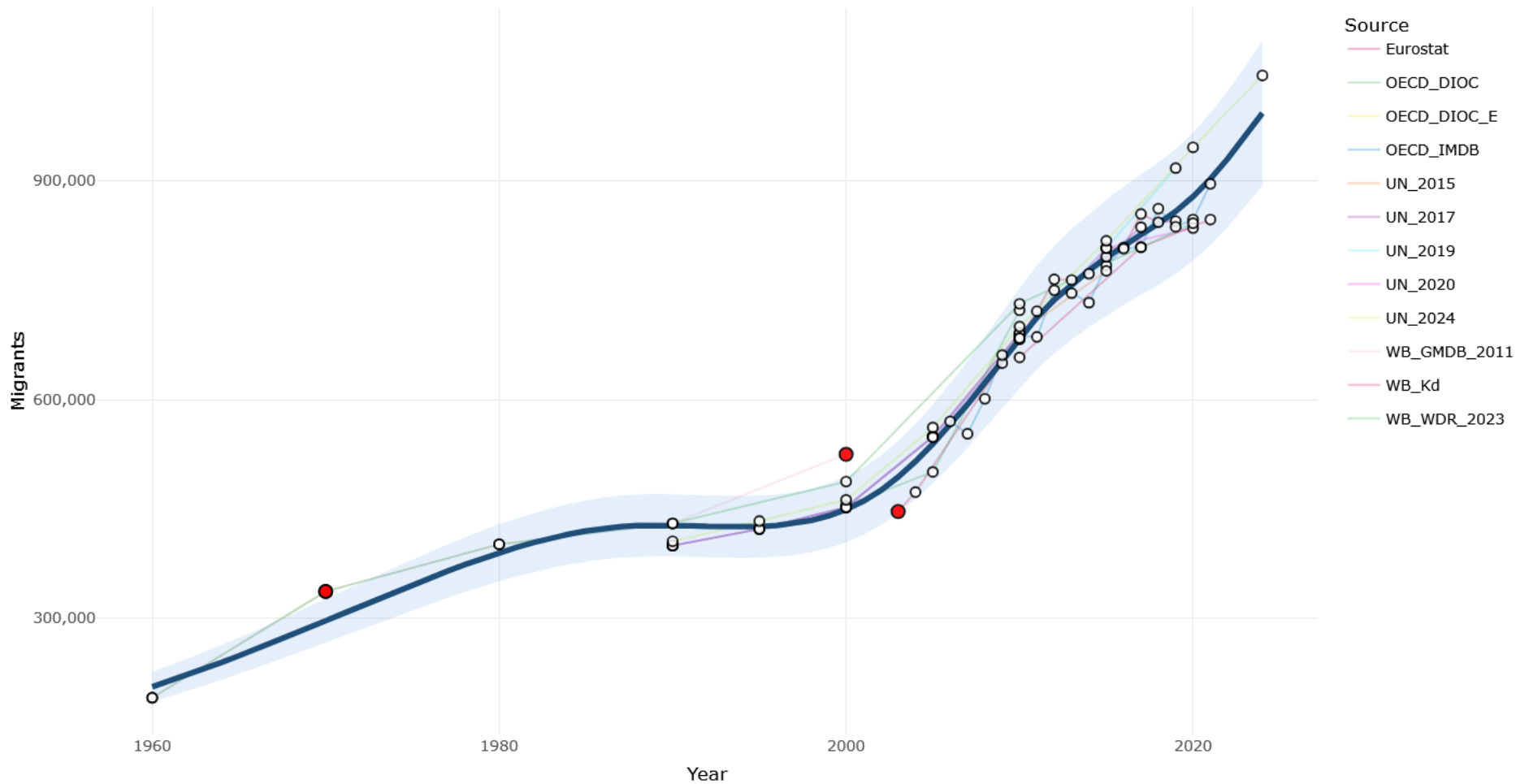
Mexicans in the USA



Romania (ROU) → Italy (ITA)



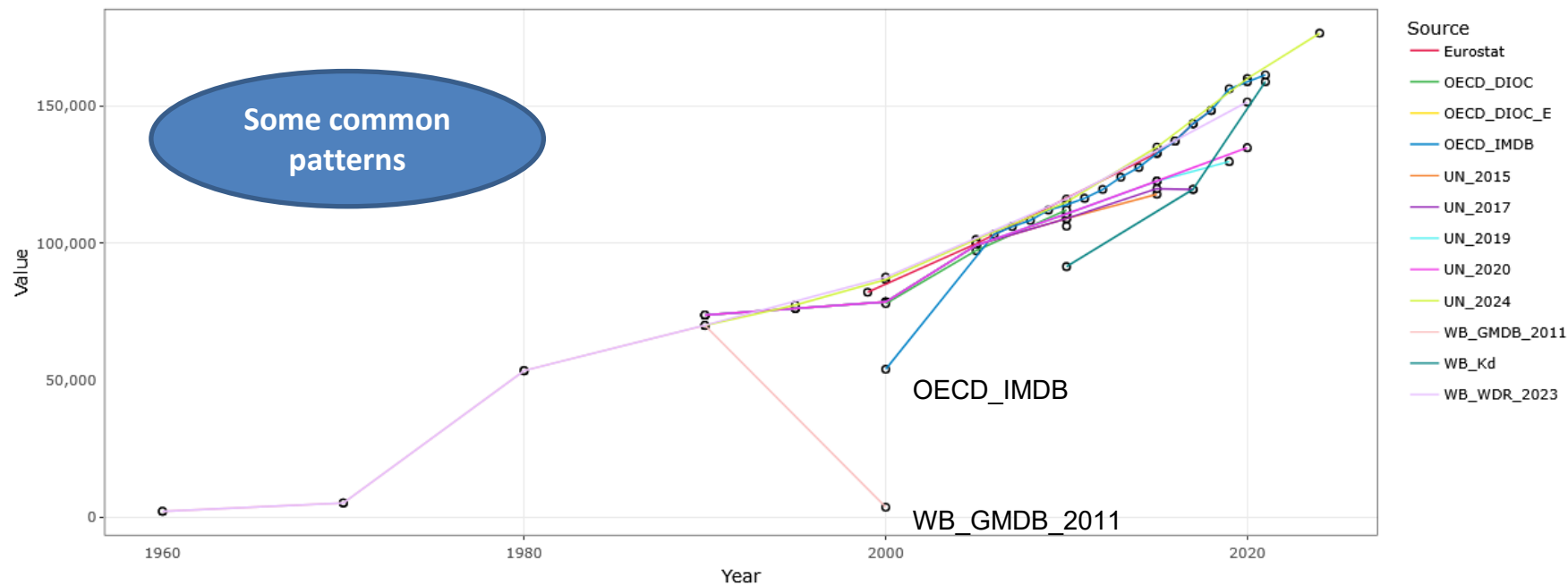
India (IND) → United Kingdom (GBR)



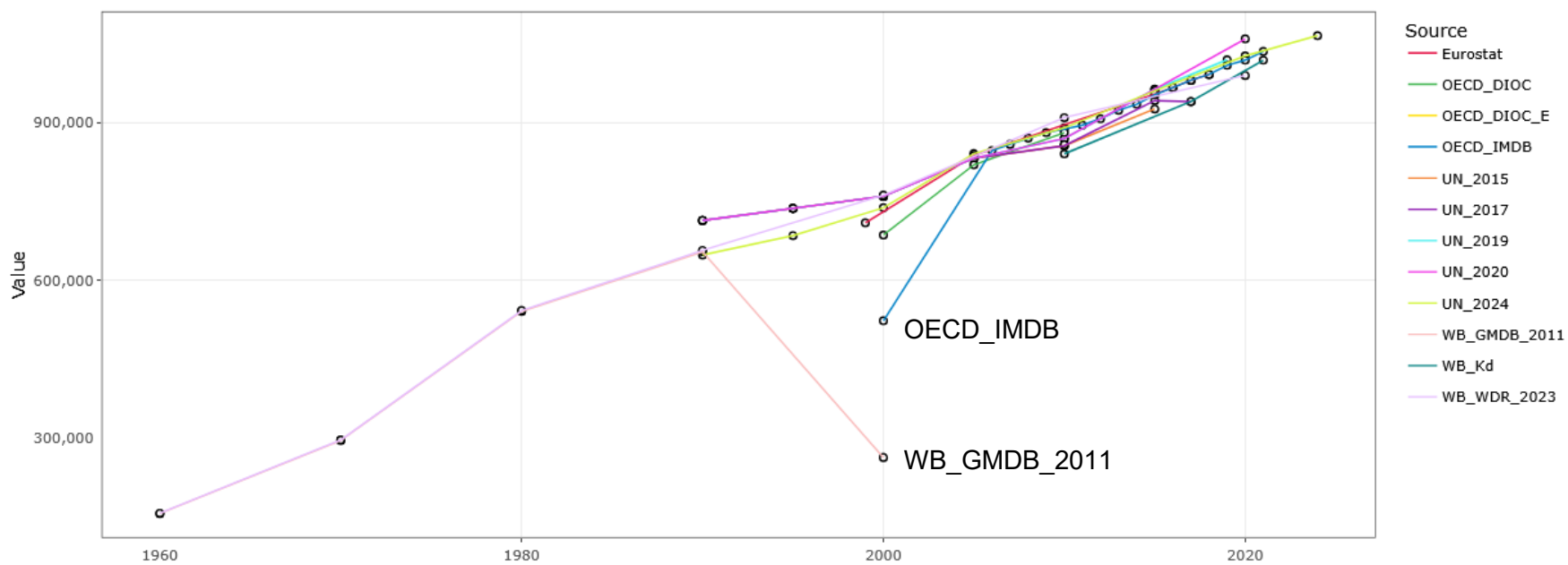
Des cas problématiques



Outlier Detection: SEN → FRA

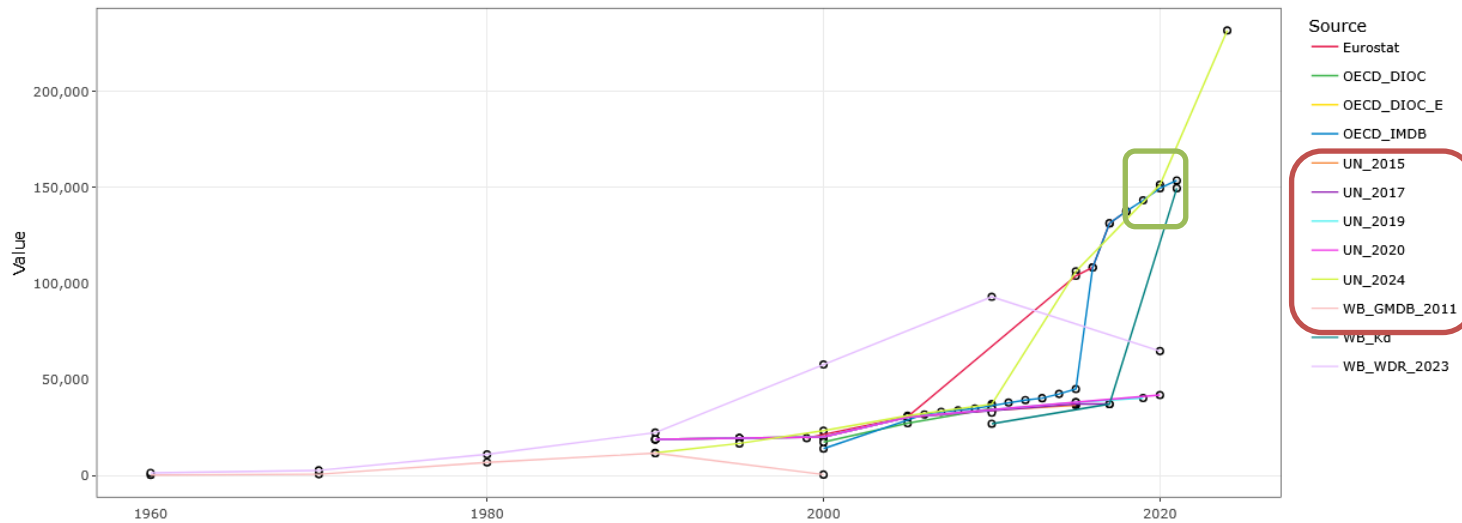


Outlier Detection: MAR → FRA

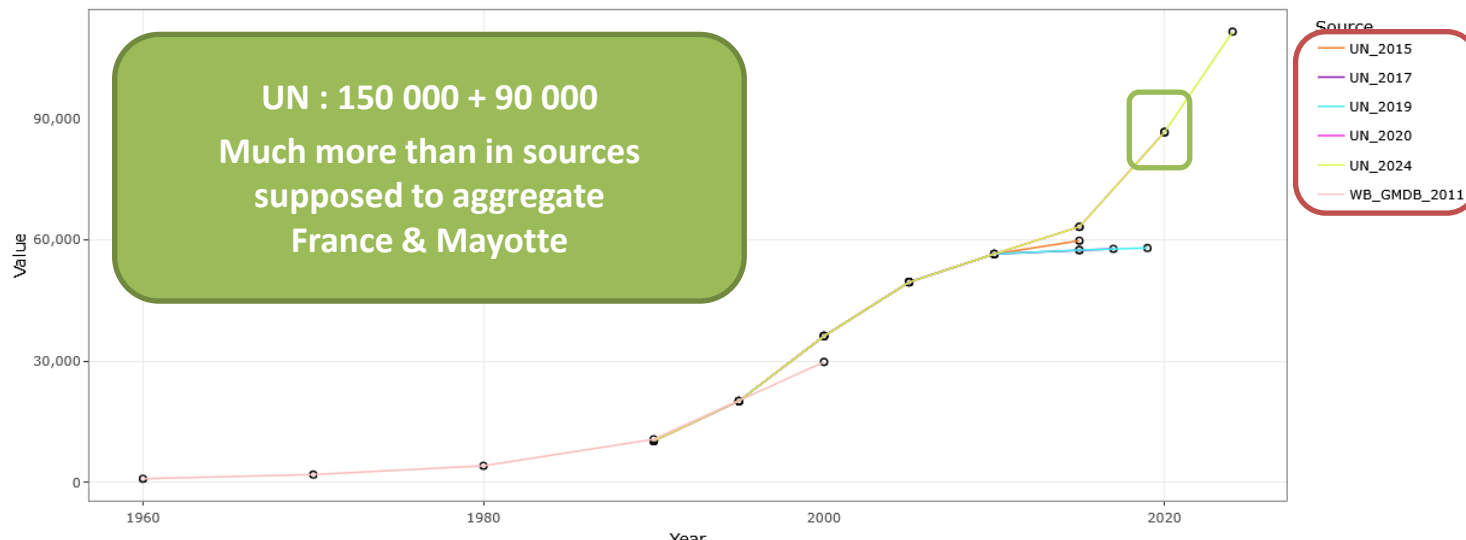


Comorians in France and in Mayotte

Outlier Detection: COM → FRA

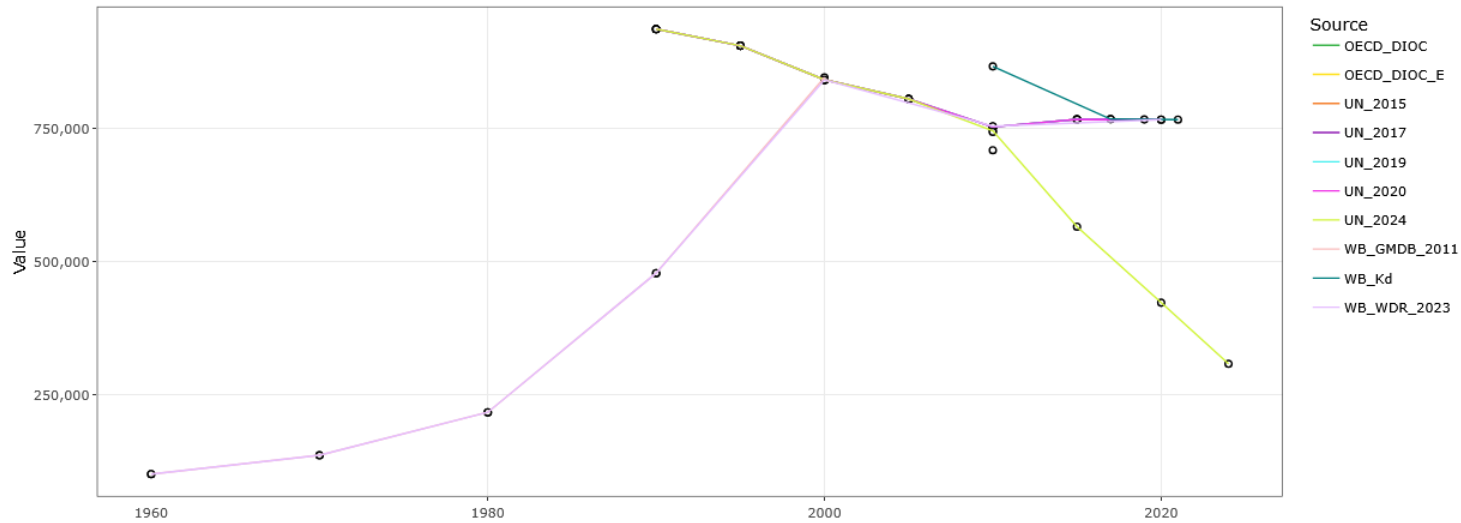


Outlier Detection: COM → MYT

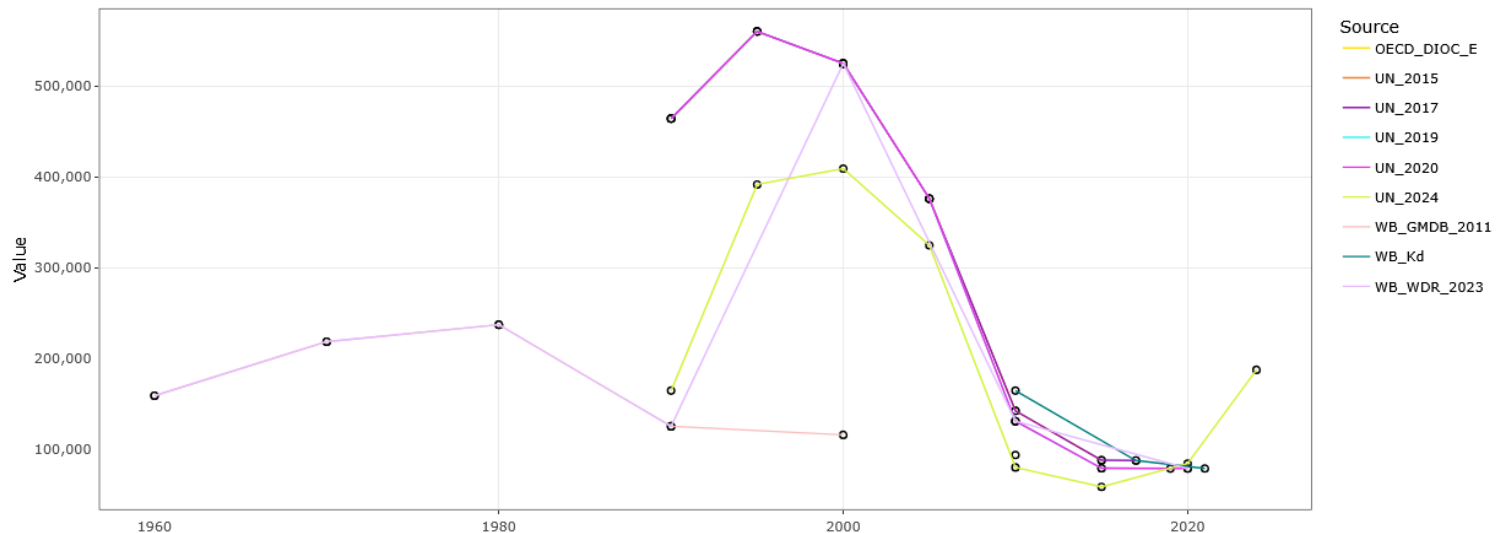


Azeri migrants in Russia and Armenia

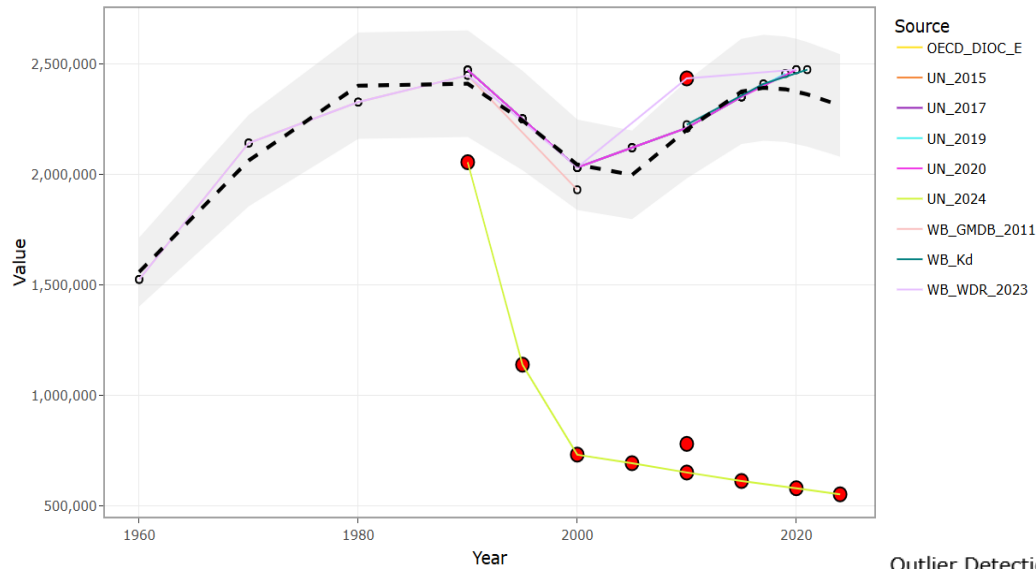
Outlier Detection: AZE $\rightarrow$ RUS



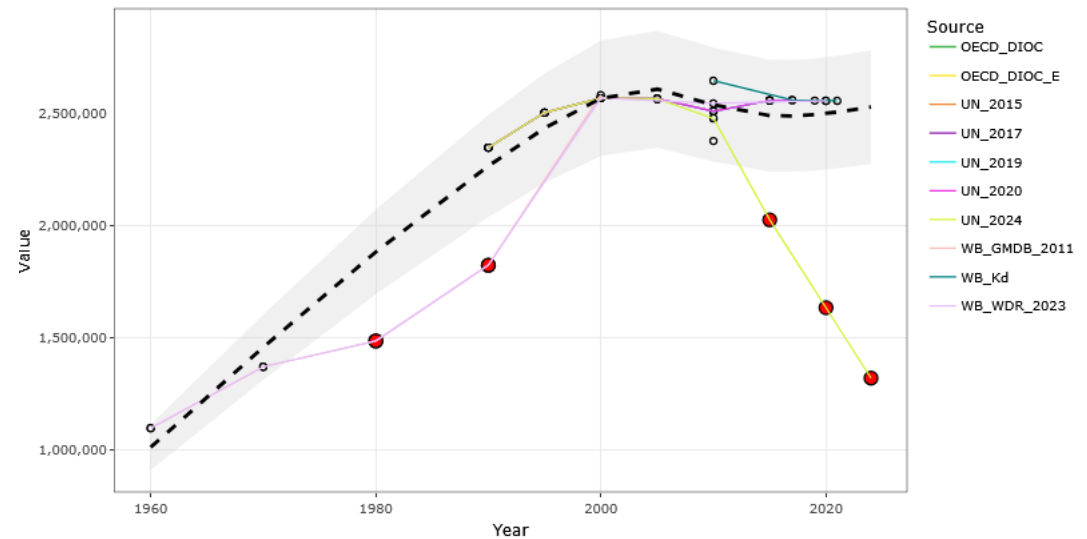
Outlier Detection: AZE $\rightarrow$ ARM



Outlier Detection: RUS → KAZ

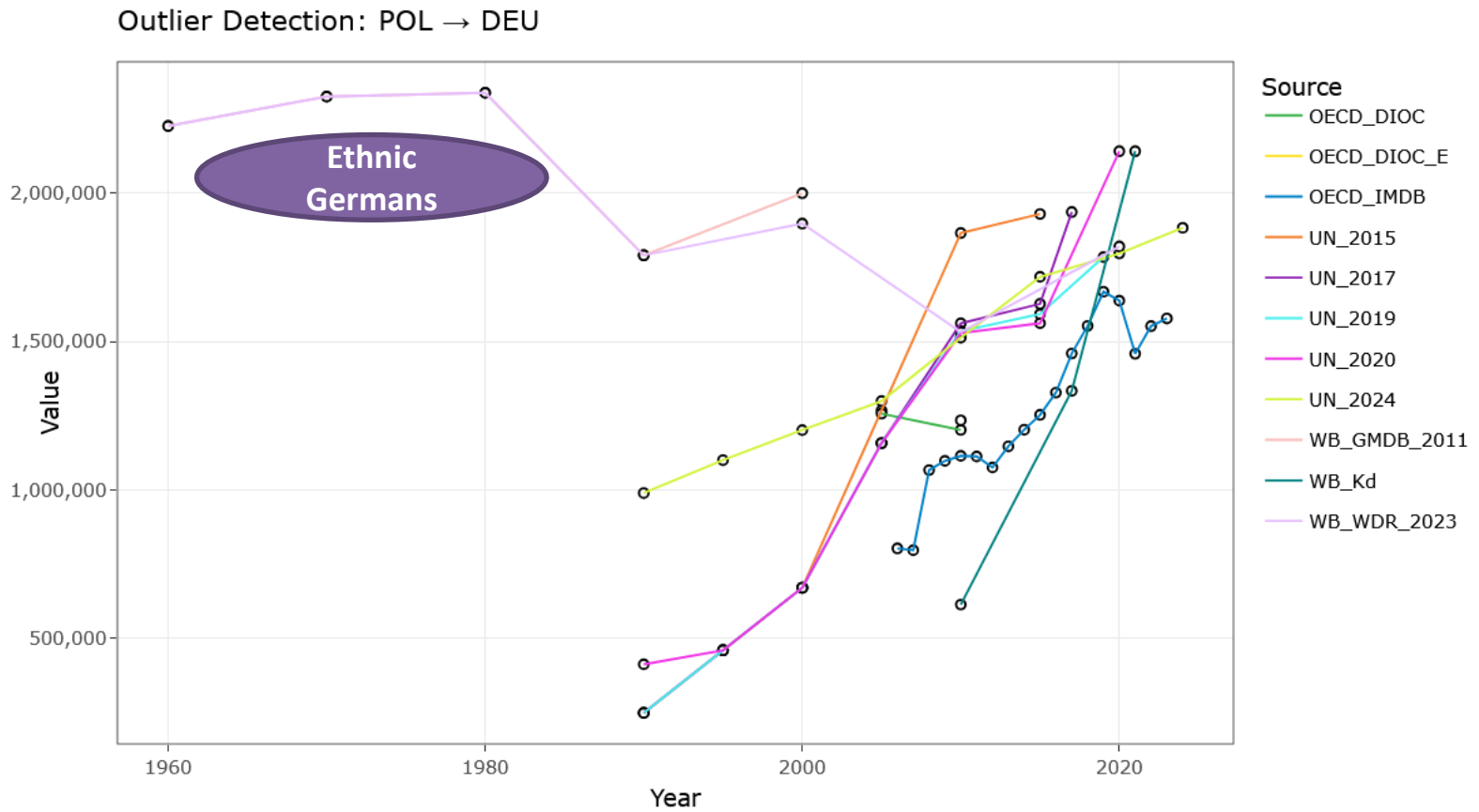


Outlier Detection: KAZ → RUS



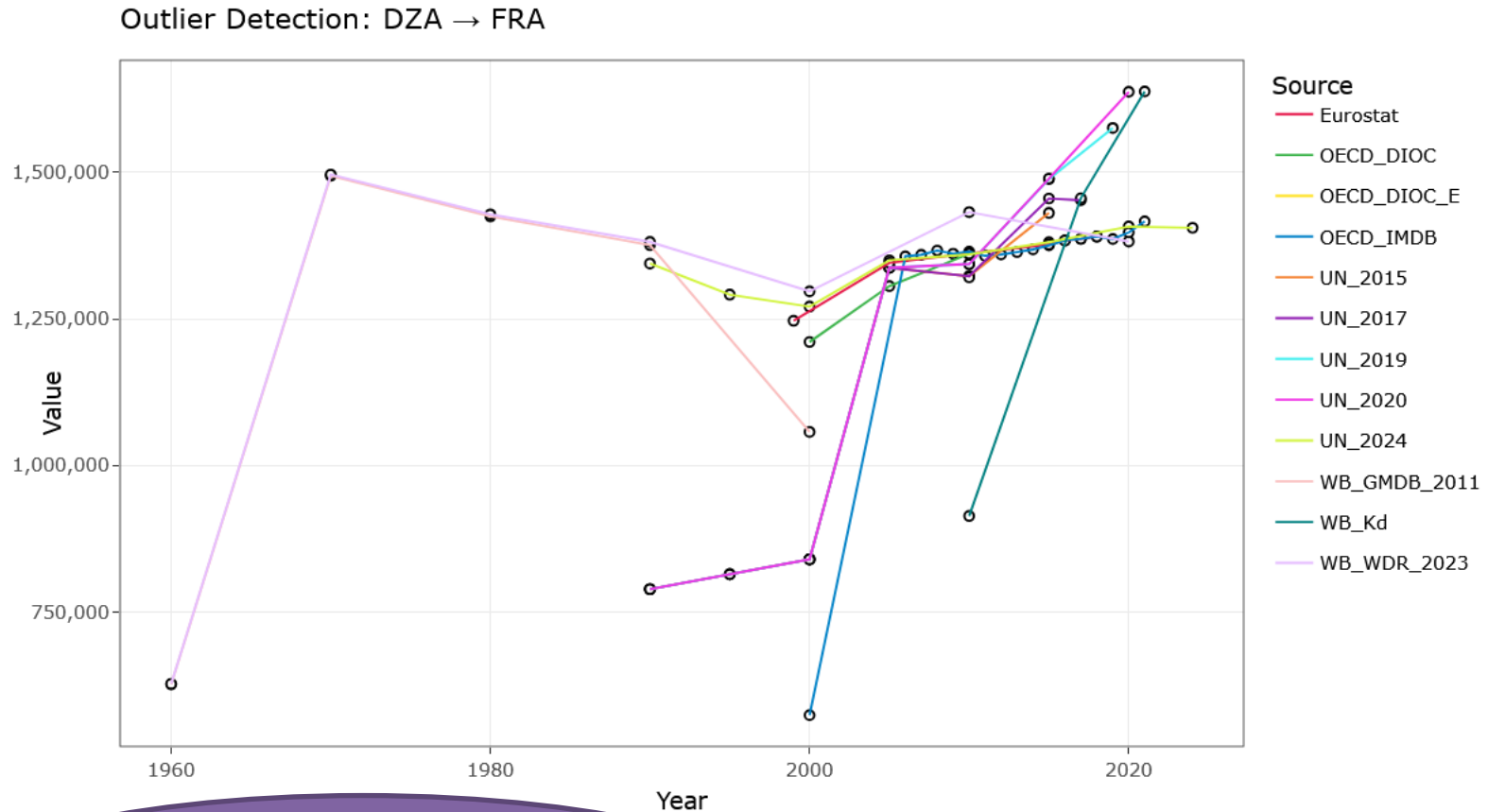
Defining migrants

Polish migrants in Germany (ex-empire effect)



Definition issues

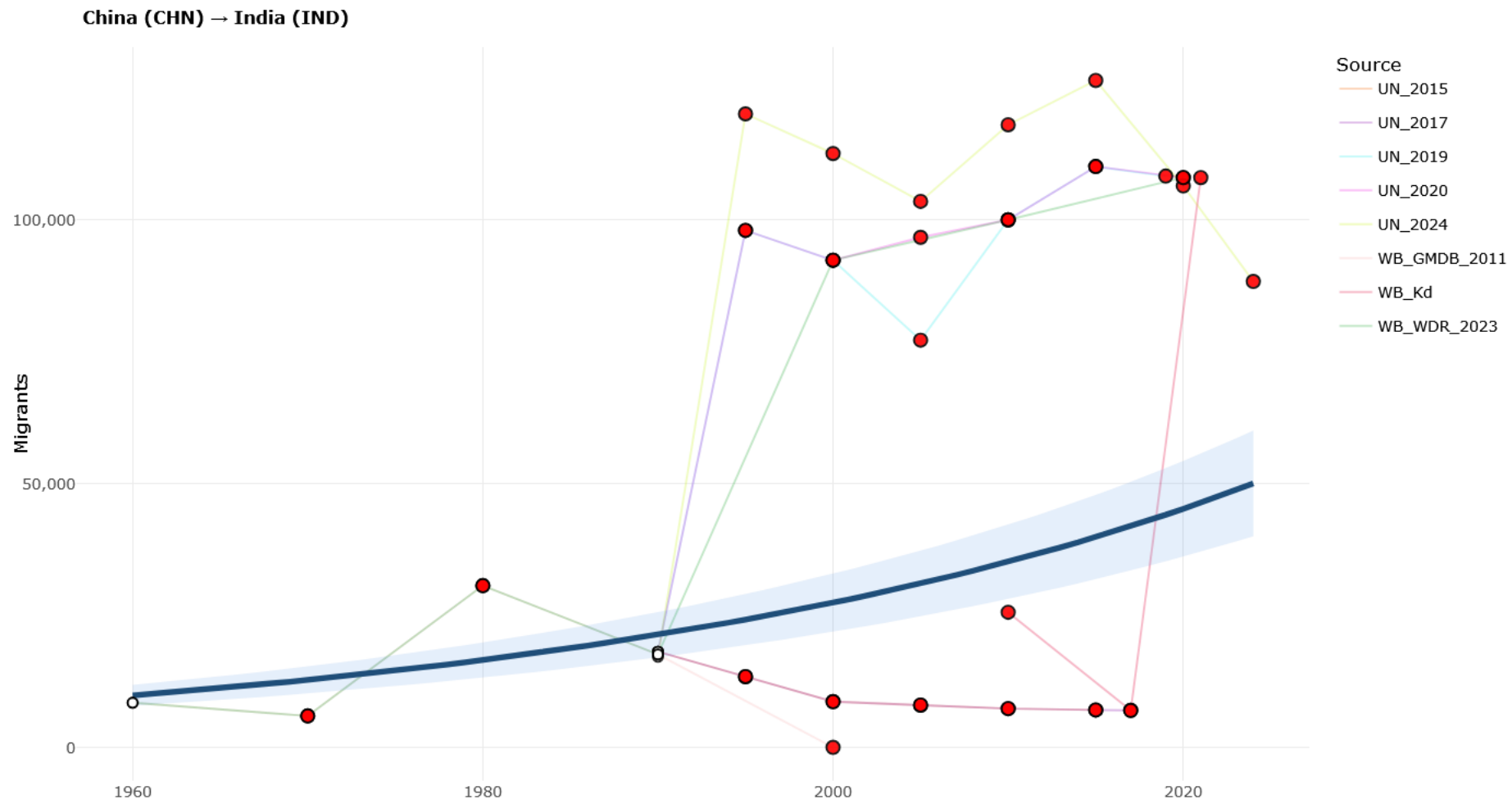
Algerian migrants in France (former colony)



With or without repatriates

- French official def. of immigrants
- Foreign born with French citizenship

Chinese migrants in India



Blue area (tolerance band): +/- 20% around the log-gam modelled spline

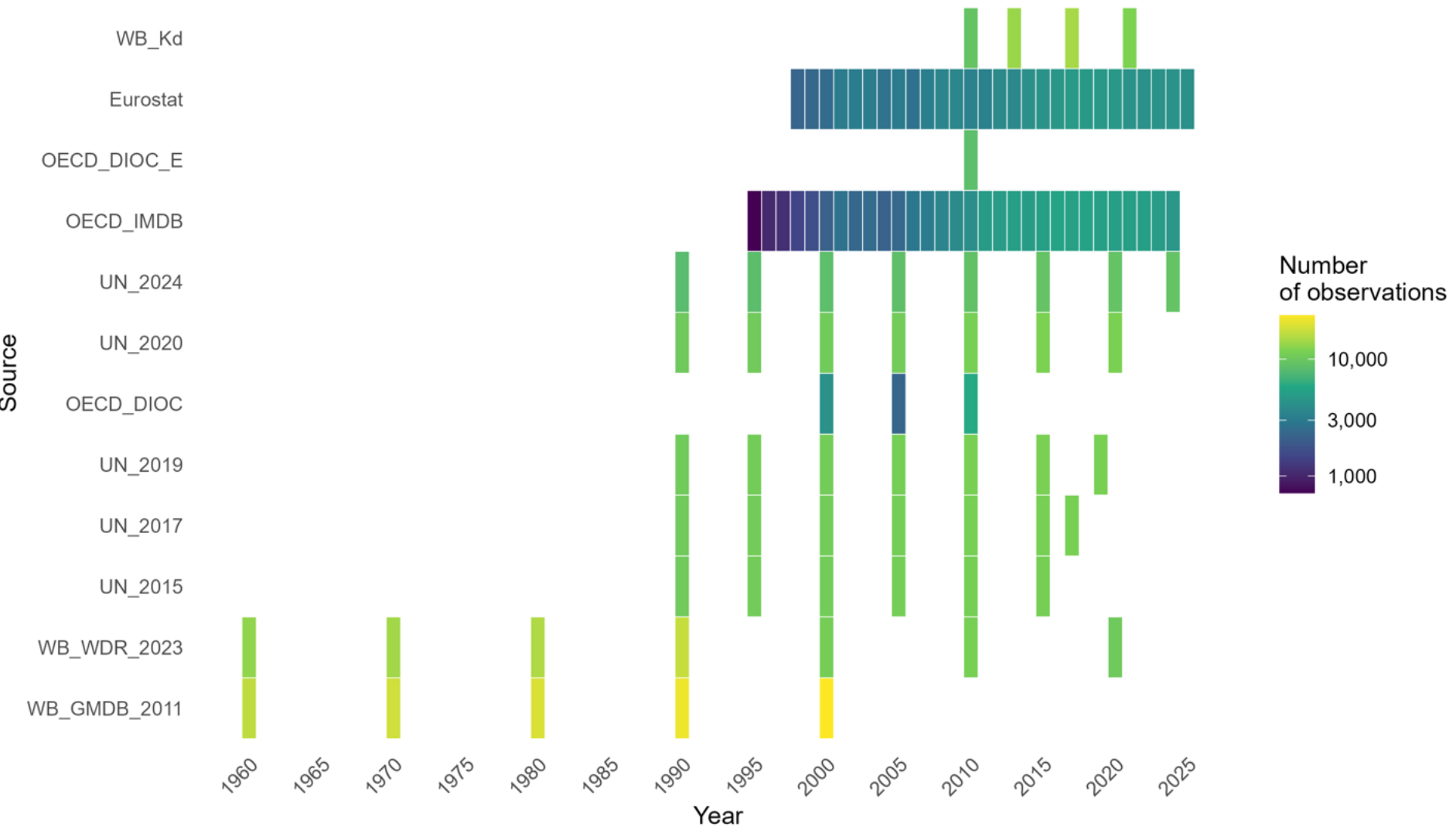
Red dots: outliers

Description des sources

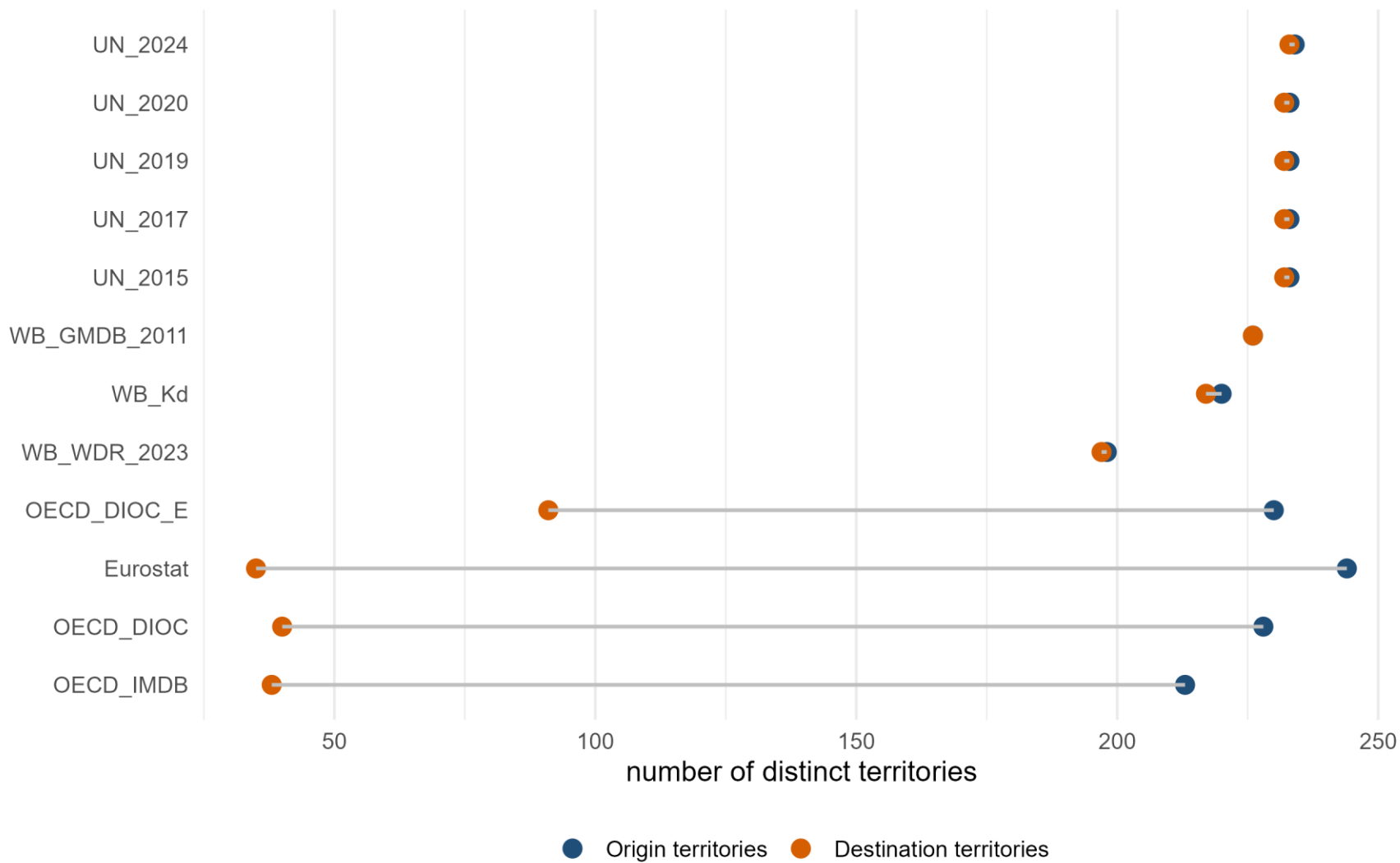


Super ! Mais revoir le gradient de couleurs

Temporal Coverage by Source

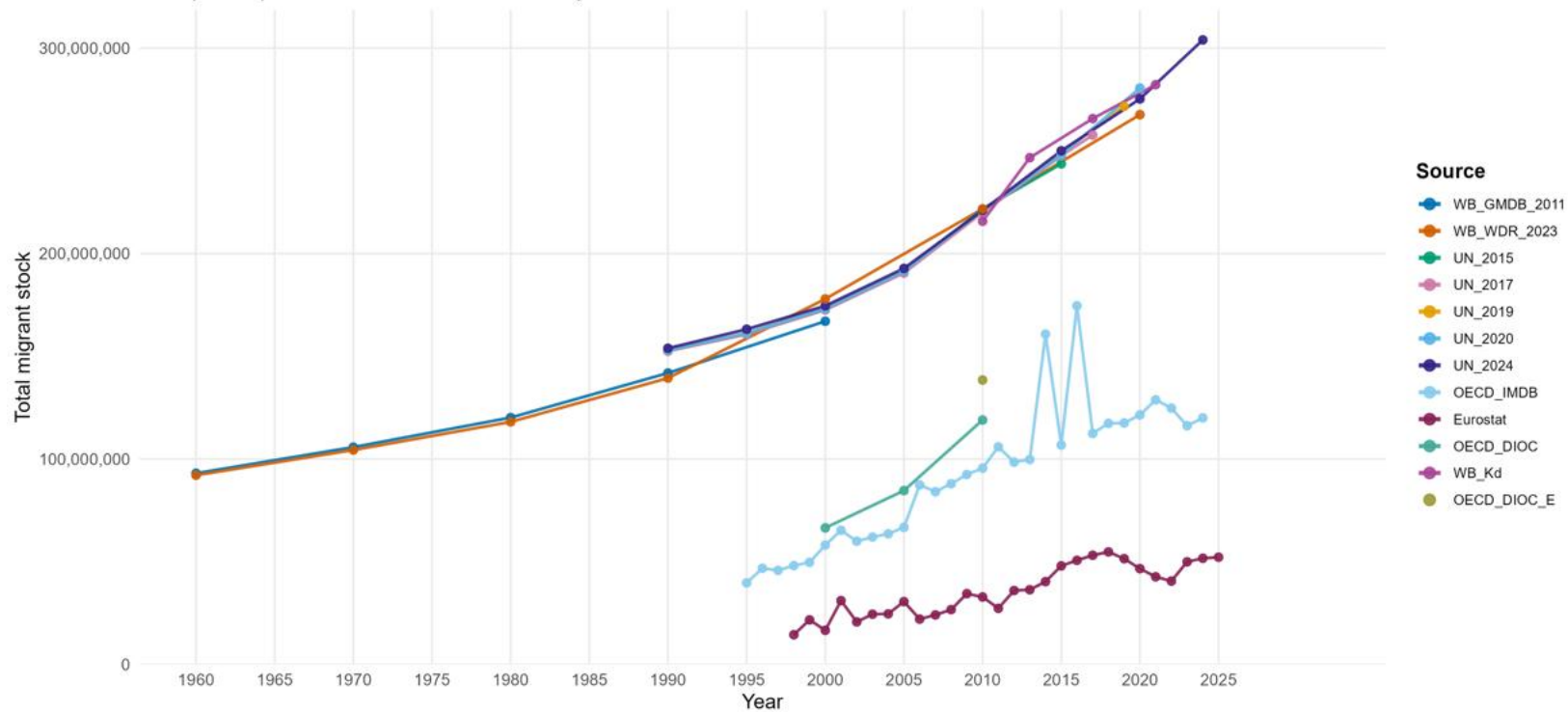


Territory Coverage by Source (FULL_MIG)



Migrant Stock over Time

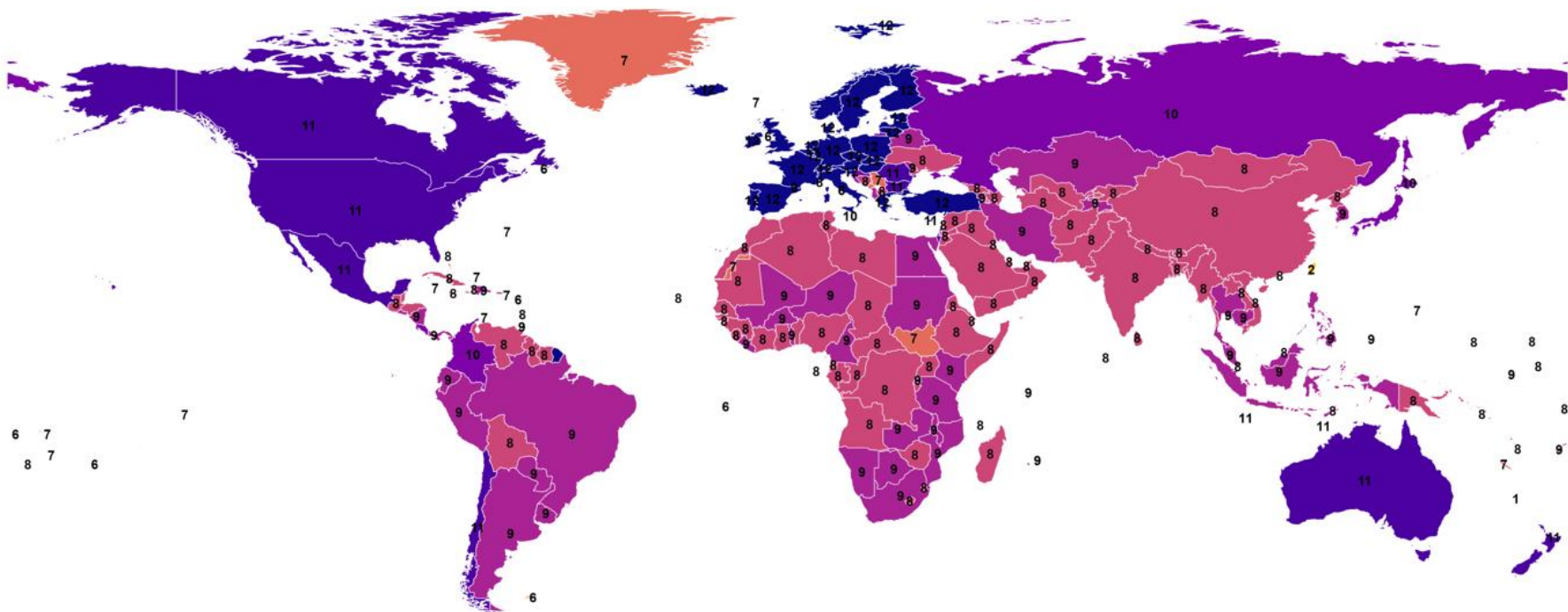
Each point represents total value for a source × year



sur la période 1960-2024 ? Ou 1990-2020 ?

Number of Sources by Destination Country

Labels: number of distinct sources

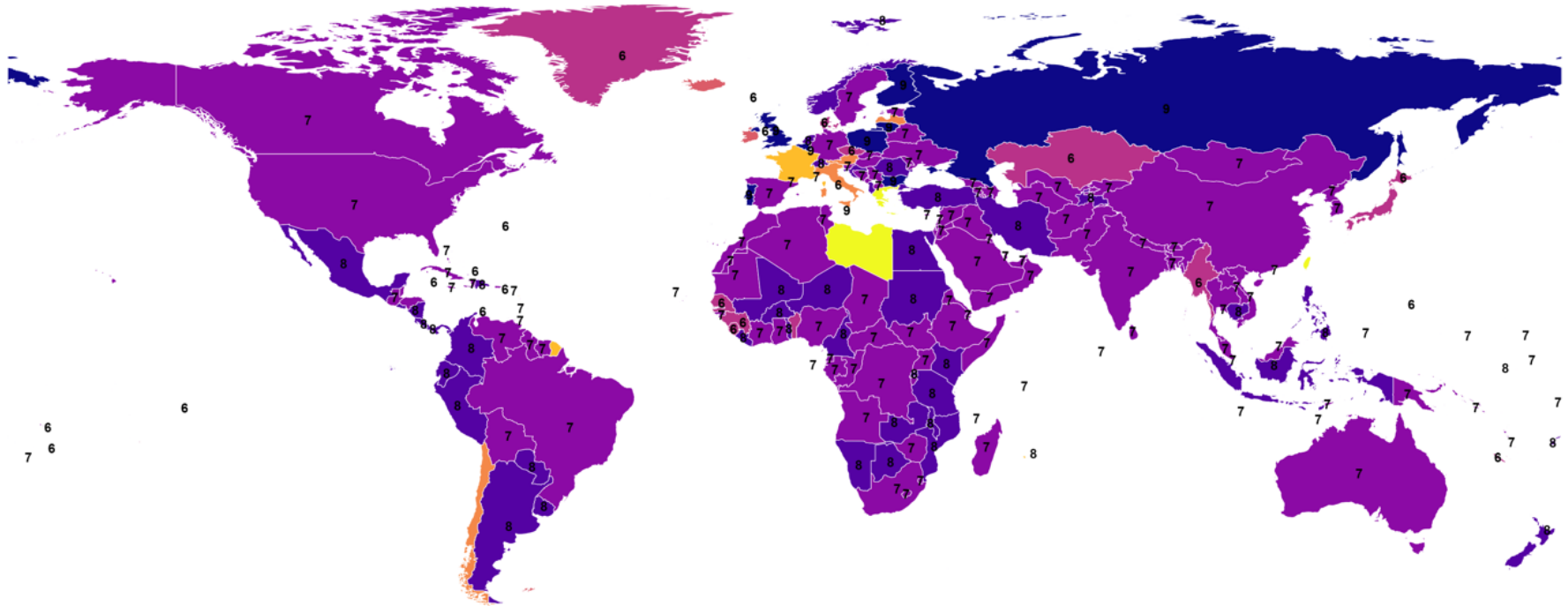




Je ne la comprends pas bien. France zarb ?

Sources per Destination — 'Other' Origin Flows

Shaded by number of distinct sources; labels for destinations with 6+ sources



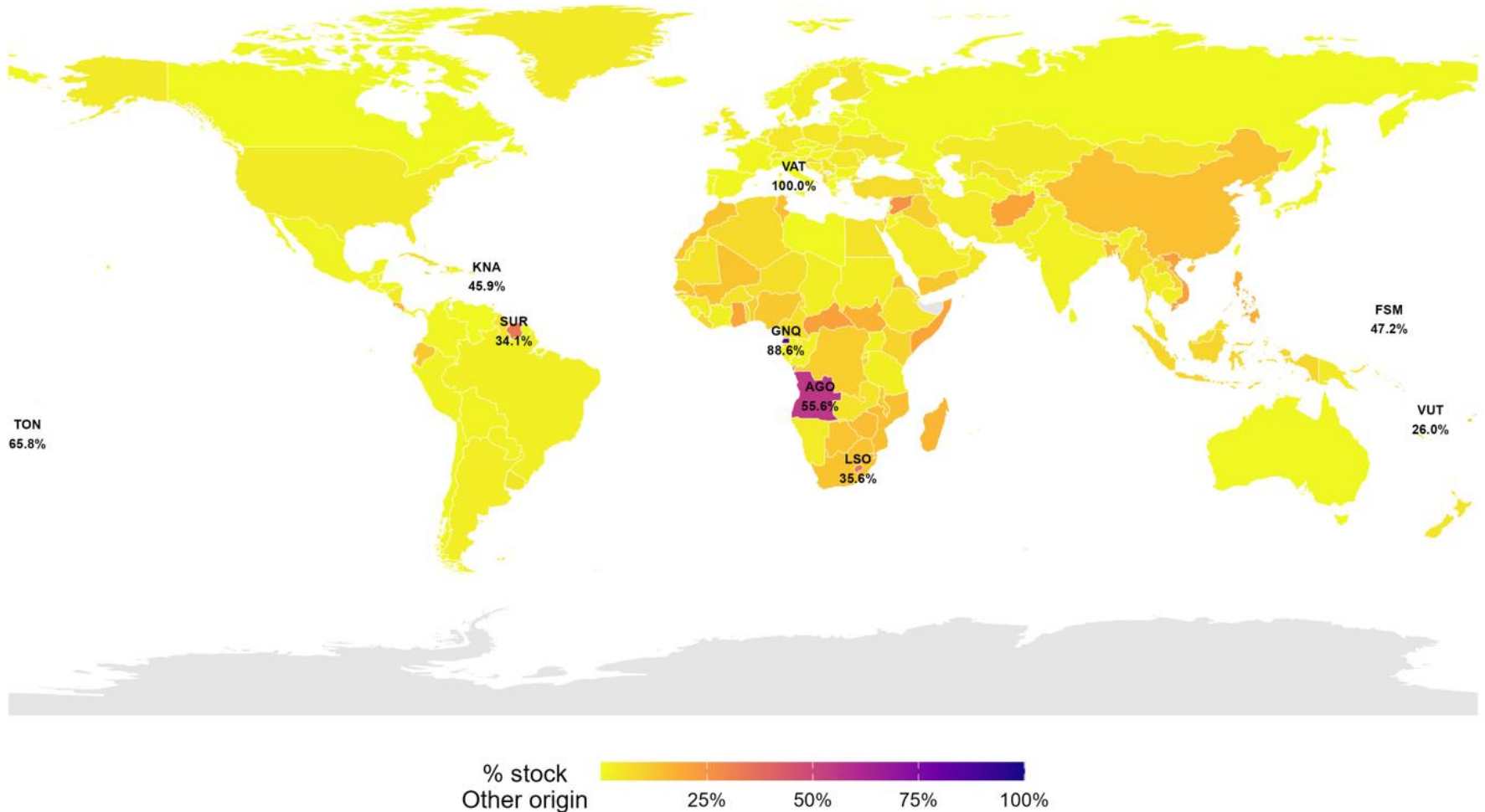
Number
of sources



Les cartes en html, ce serait super pour pouvoir voir toutes les étiquettes (nom de pays et valeurs) + pas sûr que l'échelle continue soit le meilleur choix

Destinations of Migrants with 'Other' Origin

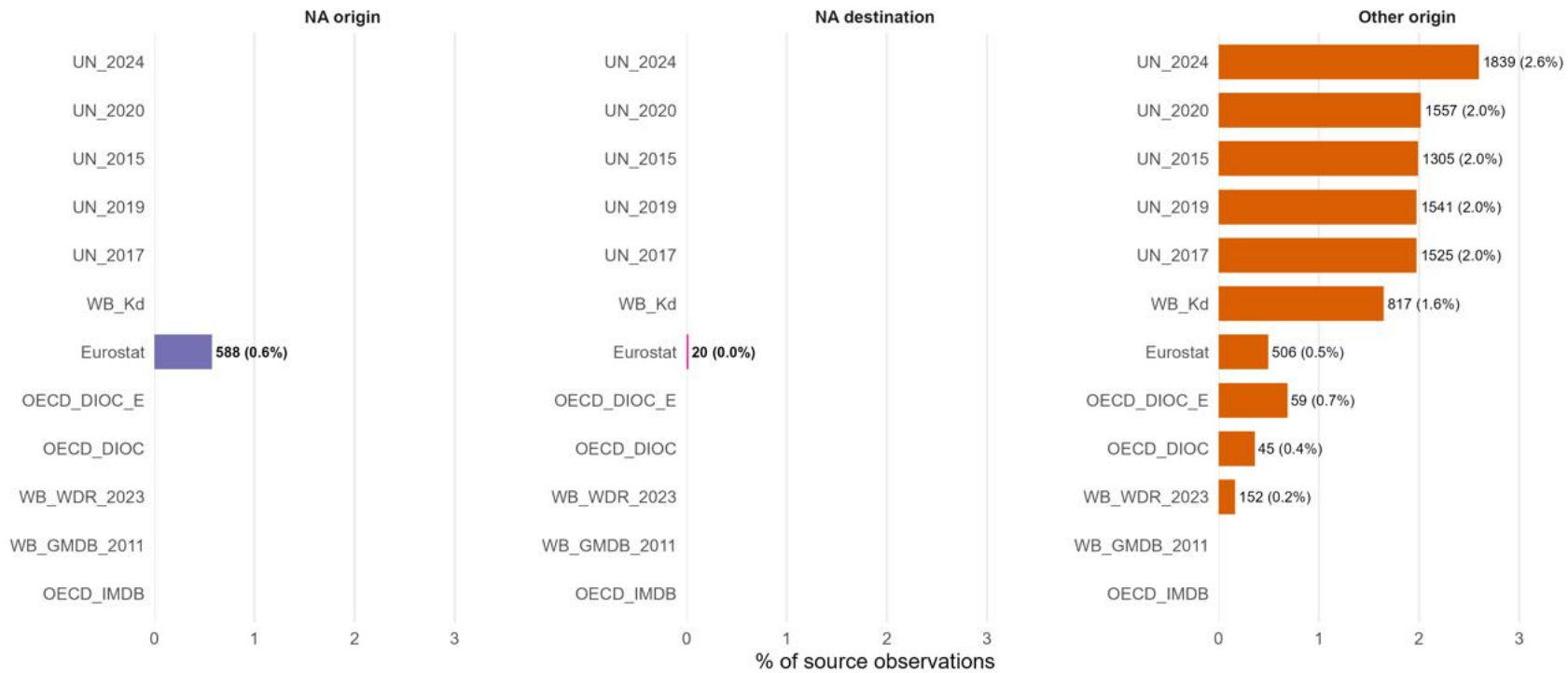
Shaded by share of destination migrant stock; top 12 percentages labelled



Que fait on des na ?

NA and Other by Source

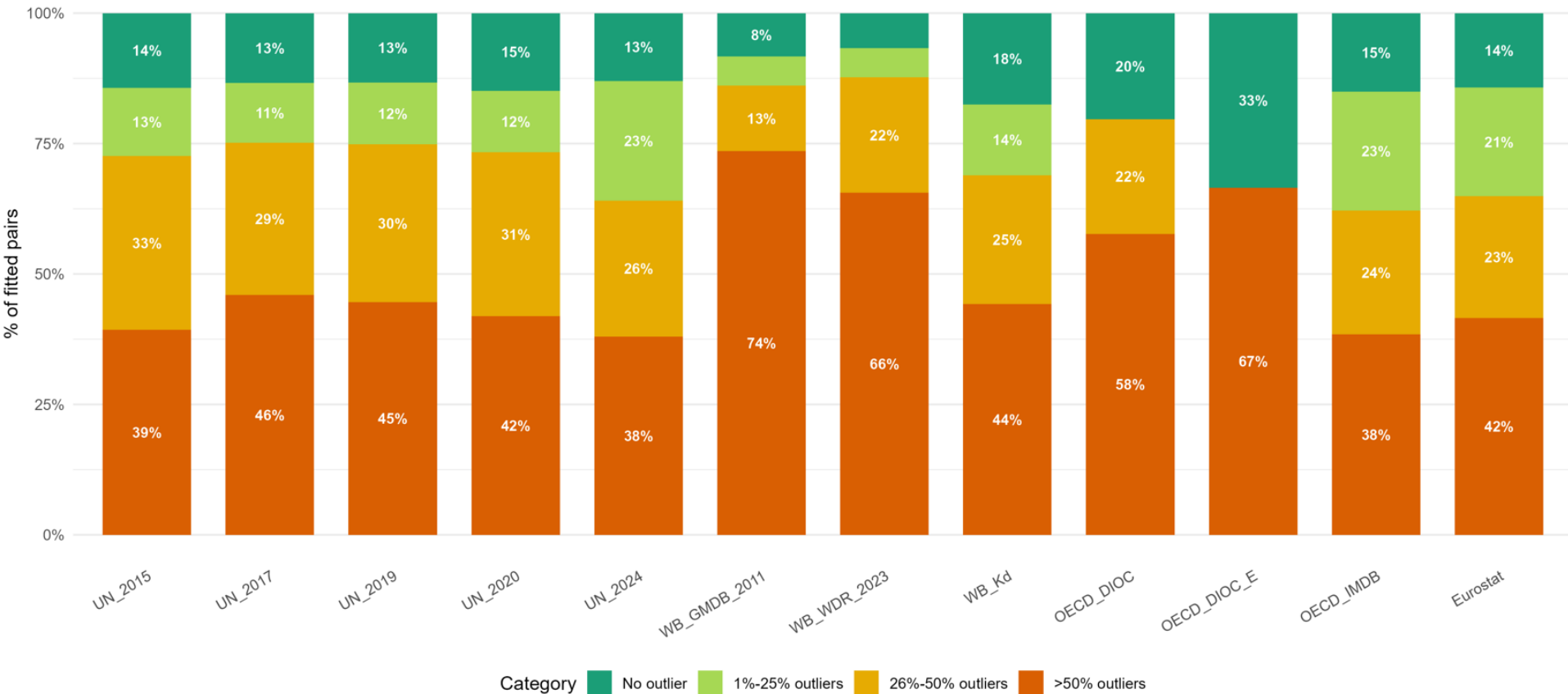
Percent of source observations



% w/o outliers étonnamment élevé. DIOC zarb !

Pair Composition by Outlier Share

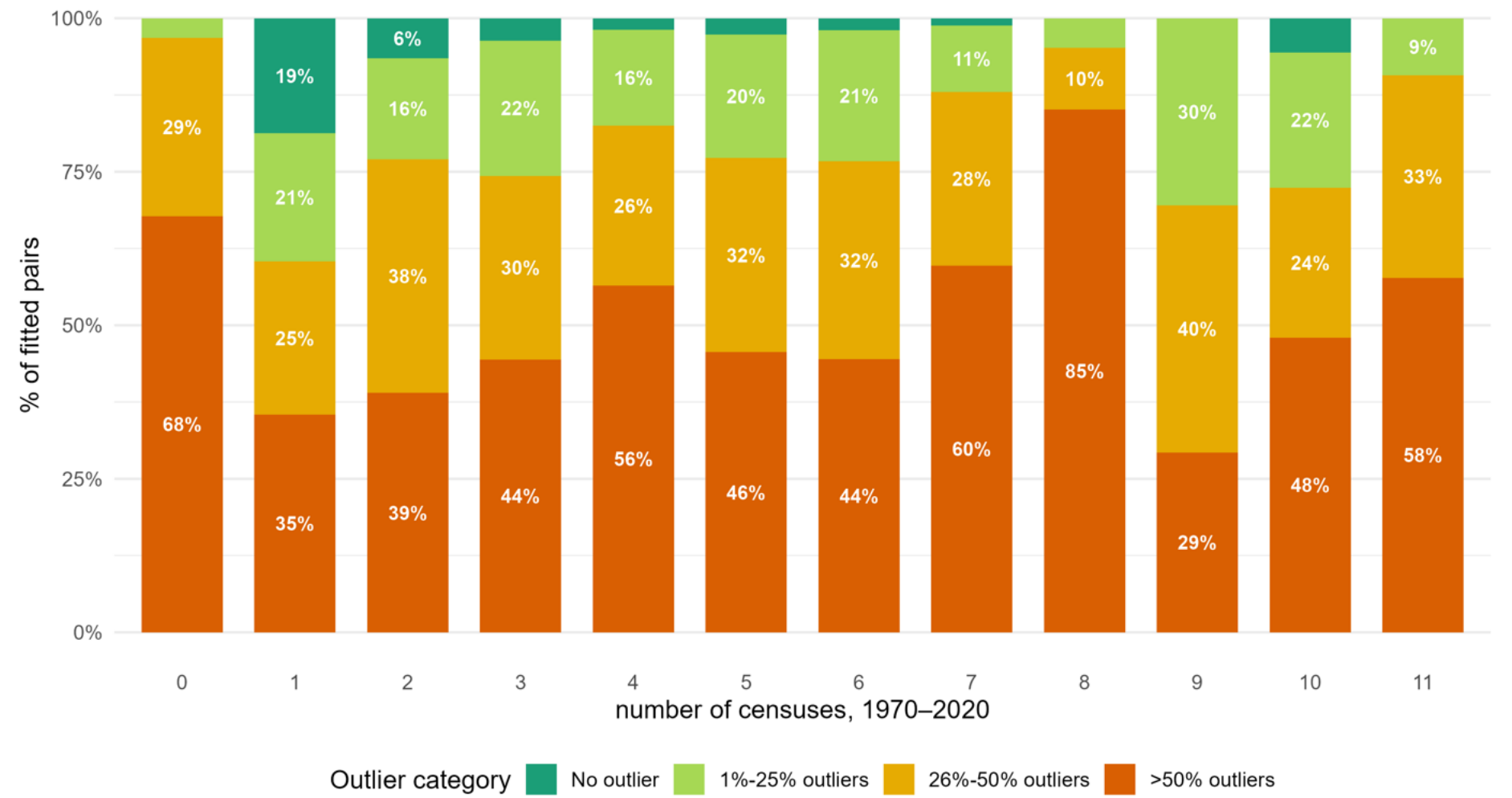
Log-GAM, $\pm 10\%$ threshold, fitted pairs only



Comment sont traités les overseas pour compter le nb de recensements ?

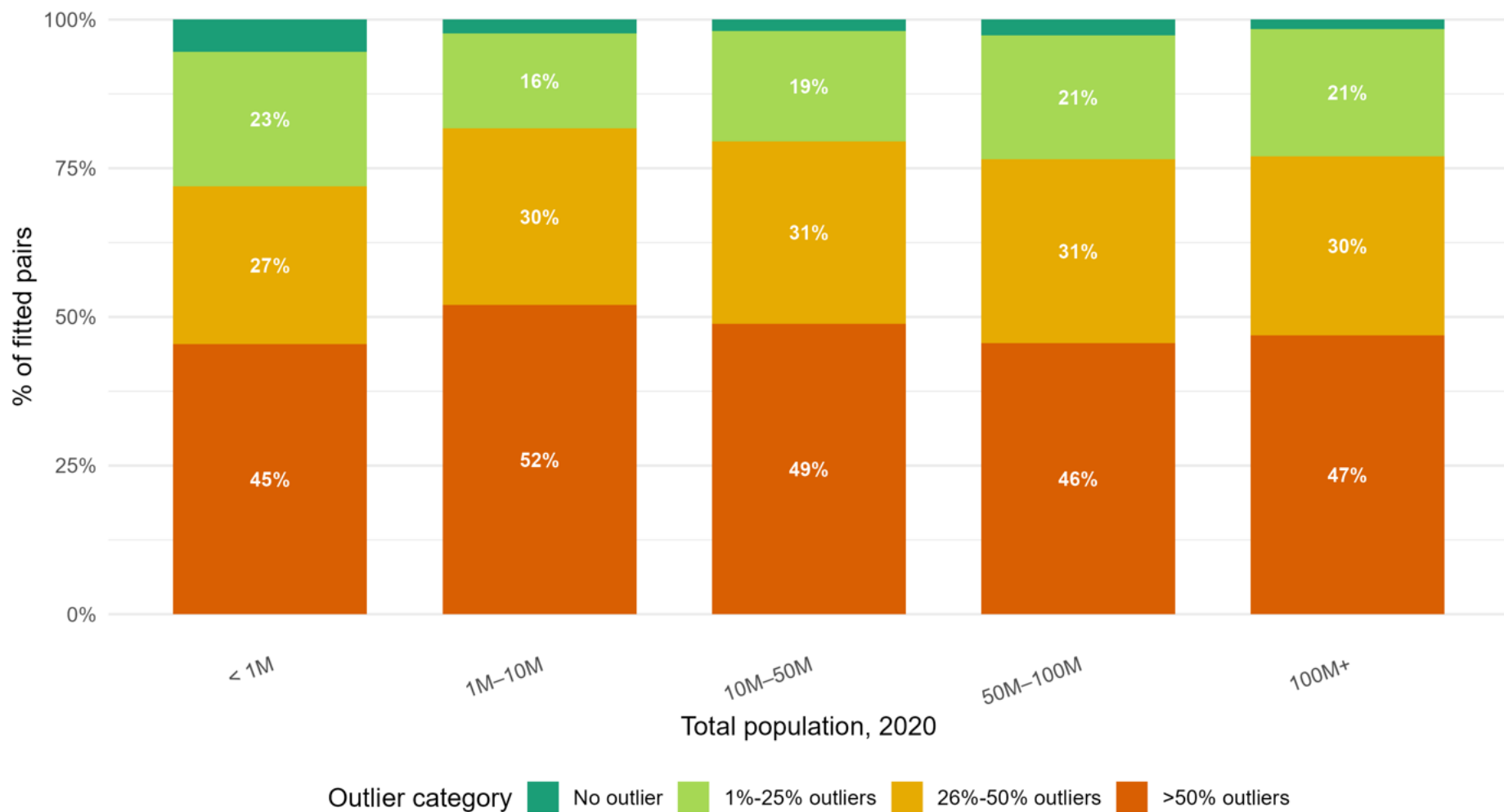
Outlier Share by Number of Censuses — Destination

exact number of censuses, 1970–2020



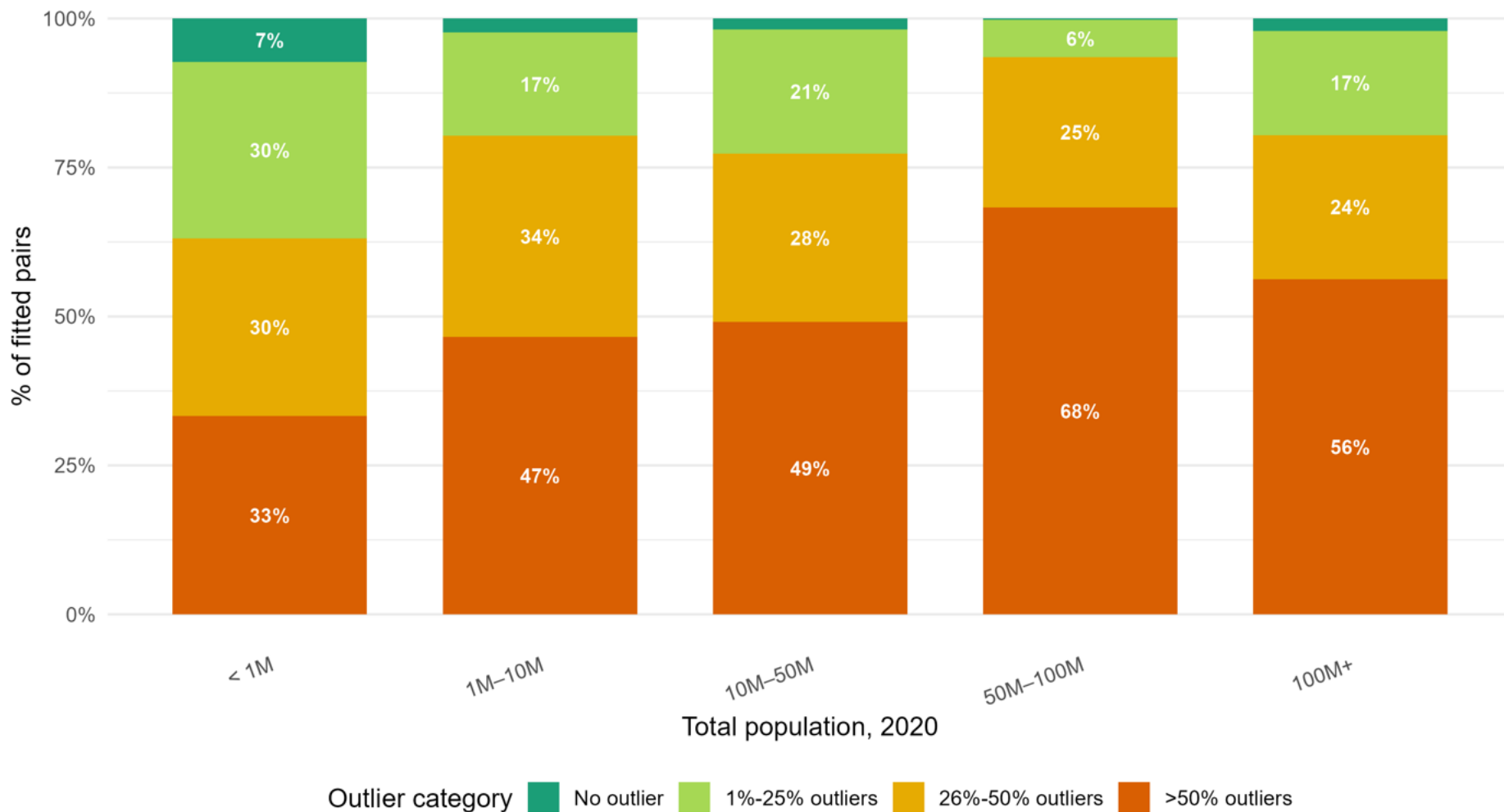
Outliers by 2020 Population - Origin

Population classes, 2020



Outliers by 2020 Population - Destination

Population classes, 2020



**Hyp: the more the number of BIMS-DB for a country-pair,
the higher the risk of discrepancies,
and the more the quantity of outliers**



Outlier Share by Number of Sources per Pair

Fitted pairs only; classes based on n_sources

